

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408917
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Khairkhahan; Alex

Methods and devices for valve clip excision

Abstract

A system for excising an implanted clip approximating opposed valve leaflets in a heart valve includes a capture catheter configured to be introduced proximate the valve leaflets on one side of the clip, a transfer catheter configured to be introduced proximate the valve leaflets on another side of the clip, and a cutting tool configured to be deployed between the capture and transfer catheters and to be engaged against tissue of at least one of the valve leaflets and to excise the clip. A removal catheter may optionally be used to remove the clip from the heart.

Inventors:	Khairkhahan; Alex (Palo Alto, CA)
Applicant:	Evalue, Inc. (Santa Clara, CA)
Family ID:	60892878
Assignee:	EVALVE, INC. (Santa Clara, CA)
Appl. No.:	16/916754
Filed:	June 30, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20200323528 A1	Oct. 15, 2020

Related U.S. Application Data

division parent-doc US 15642245 20170705 US 10736632 child-doc US 16916754
us-provisional-application US 62418571 20161107
us-provisional-application US 62359121 20160706

Publication Classification

Int. Cl.: **A61B17/10** (20060101); **A61B17/128** (20060101); **A61B17/32** (20060101);
A61B17/3205 (20060101); **A61B18/14** (20060101); A61B17/00 (20060101); A61B17/08
(20060101); A61B17/22 (20060101); A61B18/00 (20060101); A61F2/24 (20060101)

U.S. Cl.:

CPC **A61B17/10** (20130101); **A61B17/1285** (20130101); **A61B17/320016** (20130101);
A61B17/3205 (20130101); **A61B18/1492** (20130101); A61B2017/00243 (20130101);
A61B2017/00358 (20130101); A61B2017/00783 (20130101); A61B2017/00876
(20130101); A61B17/083 (20130101); A61B2017/22035 (20130101); A61B2017/32006
(20130101); A61B2018/00369 (20130101); A61B2018/00601 (20130101);
A61B2018/1407 (20130101); A61B2018/144 (20130101); A61F2/246 (20130101);
A61F2/2466 (20130101)

Field of Classification Search

CPC: A61B (17/320016); A61B (17/3205); A61B (17/10); A61B (17/1285); A61B (17/083);
A61B (2017/00243); A61B (2017/00783); A61B (2017/32006); A61B (2017/22035);
A61B (2017/00876); A61B (2017/00358); A61B (18/1492); A61B (2018/144); A61B
(2018/1407); A61B (2018/00601); A61B (2018/00369); A61F (2/2466); A61F (2/246)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1996261	12/1934	Storz	N/A	N/A
2097018	12/1936	Chamberlin	N/A	N/A
2108206	12/1937	Meeker	N/A	N/A
3296668	12/1966	Aiken	N/A	N/A
3378010	12/1967	Codling et al.	N/A	N/A
3470875	12/1968	Johnson et al.	N/A	N/A
3557780	12/1970	Sato	N/A	N/A
3671979	12/1971	Moulopoulos	N/A	N/A
3675639	12/1971	Cimber	N/A	N/A
3776237	12/1972	Hill et al.	N/A	N/A
3874338	12/1974	Happel	N/A	N/A
3874388	12/1974	King et al.	N/A	N/A
4007743	12/1976	Blake	N/A	N/A
4056854	12/1976	Boretos et al.	N/A	N/A
4064881	12/1976	Meredith	N/A	N/A
4091815	12/1977	Larsen	N/A	N/A
4112951	12/1977	Hulka et al.	N/A	N/A
4235238	12/1979	Ogiu et al.	N/A	N/A
4297749	12/1980	Davis et al.	N/A	N/A
4312337	12/1981	Donahue	N/A	N/A
4425908	12/1983	Simon	N/A	N/A
4458682	12/1983	Cerwin	N/A	N/A
4484579	12/1983	Meno et al.	N/A	N/A
4487205	12/1983	Di Giovanni et al.	N/A	N/A

4498476	12/1984	Cerwin et al.	N/A	N/A
4510934	12/1984	Batra	N/A	N/A
4531522	12/1984	Bedi et al.	N/A	N/A
4578061	12/1985	Lemelson	N/A	N/A
4641366	12/1986	Yokoyama et al.	N/A	N/A
4686965	12/1986	Bonnet et al.	N/A	N/A
4777951	12/1987	Cribier et al.	N/A	N/A
4809695	12/1988	Gwathmey et al.	N/A	N/A
4872455	12/1988	Pinchuk et al.	N/A	N/A
4878495	12/1988	Grayzel	N/A	N/A
4917089	12/1989	Sideris	N/A	N/A
4944295	12/1989	Gwathmey et al.	N/A	N/A
4969890	12/1989	Sugita et al.	N/A	N/A
4994077	12/1990	Dobben	N/A	N/A
5015249	12/1990	Nakao et al.	N/A	N/A
5019096	12/1990	Fox et al.	N/A	N/A
5042707	12/1990	Taheri	N/A	N/A
5047041	12/1990	Samuels	N/A	N/A
5049153	12/1990	Nakao et al.	N/A	N/A
5053043	12/1990	Gottesman et al.	N/A	N/A
5061277	12/1990	Carpentier et al.	N/A	N/A
5069679	12/1990	Taheri	N/A	N/A
5071428	12/1990	Chin	N/A	N/A
5078722	12/1991	Stevens	N/A	N/A
5078723	12/1991	Dance et al.	N/A	N/A
5108368	12/1991	Hammerslag et al.	N/A	N/A
5125758	12/1991	Dewan	N/A	N/A
5171252	12/1991	Friedland	N/A	N/A
5171259	12/1991	Inoue	N/A	N/A
5190554	12/1992	Coddington et al.	N/A	N/A
5195968	12/1992	Lundquist et al.	N/A	N/A
5209756	12/1992	Seedhom et al.	N/A	N/A
5217460	12/1992	Knoepfler	N/A	N/A
5226429	12/1992	Kuzmak	N/A	N/A
5226911	12/1992	Chee et al.	N/A	N/A
5234437	12/1992	Sepetka	N/A	N/A
5242456	12/1992	Nash et al.	N/A	N/A
5250071	12/1992	Palermo	N/A	N/A
5251611	12/1992	Zehel et al.	N/A	N/A
5254130	12/1992	Poncet et al.	N/A	N/A
5261916	12/1992	Engelson	N/A	N/A
5271381	12/1992	Ailinger et al.	N/A	N/A
5275578	12/1993	Adams	N/A	N/A
5282845	12/1993	Bush et al.	N/A	N/A
5304131	12/1993	Paskar	N/A	N/A
5306283	12/1993	Connors	N/A	N/A
5306286	12/1993	Stack et al.	N/A	N/A
5312415	12/1993	Palermo	N/A	N/A
5314424	12/1993	Nicholas	N/A	N/A
5318525	12/1993	West et al.	N/A	N/A

5320632	12/1993	Heidmueller	N/A	N/A
5325845	12/1993	Adair	N/A	N/A
5330442	12/1993	Green et al.	N/A	N/A
5332402	12/1993	Teitelbaum	N/A	N/A
5336227	12/1993	Nakao et al.	N/A	N/A
5342393	12/1993	Stack	N/A	N/A
5350397	12/1993	Palermo et al.	N/A	N/A
5350399	12/1993	Erlebacher et al.	N/A	N/A
5359994	12/1993	Krauter et al.	N/A	N/A
5368564	12/1993	Savage	N/A	N/A
5368601	12/1993	Sauer et al.	N/A	N/A
5373854	12/1993	Kolozsi	N/A	N/A
5383886	12/1994	Kensey et al.	N/A	N/A
5387219	12/1994	Rappe	N/A	N/A
5391182	12/1994	Chin	N/A	N/A
5395030	12/1994	Kuramoto et al.	N/A	N/A
5403312	12/1994	Yates et al.	N/A	N/A
5403326	12/1994	Harrison et al.	N/A	N/A
5411552	12/1994	Andersen et al.	N/A	N/A
5417684	12/1994	Jackson et al.	N/A	N/A
5417699	12/1994	Klein et al.	N/A	N/A
5417700	12/1994	Egan	N/A	N/A
5423830	12/1994	Schneebaum et al.	N/A	N/A
5423857	12/1994	Rosenman et al.	N/A	N/A
5423858	12/1994	Bolanos et al.	N/A	N/A
5423882	12/1994	Jackman et al.	N/A	N/A
5431666	12/1994	Sauer et al.	N/A	N/A
5437551	12/1994	Chalifoux	N/A	N/A
5437681	12/1994	Meade et al.	N/A	N/A
5447966	12/1994	Hermes et al.	N/A	N/A
5450860	12/1994	O'Connor	N/A	N/A
5456400	12/1994	Shichman et al.	N/A	N/A
5456684	12/1994	Schmidt et al.	N/A	N/A
5462527	12/1994	Stevens-Wright et al.	N/A	N/A
5472044	12/1994	Hall et al.	N/A	N/A
5472423	12/1994	Gronauer	N/A	N/A
5476470	12/1994	Fitzgibbons, Jr.	N/A	N/A
5477856	12/1994	Lundquist	N/A	N/A
5478309	12/1994	Sweezer et al.	N/A	N/A
5478353	12/1994	Yoon	N/A	N/A
5487746	12/1995	Yu et al.	N/A	N/A
5496332	12/1995	Sierra et al.	N/A	N/A
5507725	12/1995	Savage et al.	N/A	N/A
5507755	12/1995	Gresl et al.	N/A	N/A
5507757	12/1995	Sauer et al.	N/A	N/A
5520701	12/1995	Lerch	N/A	N/A
5522873	12/1995	Jackman et al.	N/A	N/A
5527313	12/1995	Scott et al.	N/A	N/A
5527321	12/1995	Hinchliffe	N/A	N/A

5527322	12/1995	Klein et al.	N/A	N/A
5536251	12/1995	Evard et al.	N/A	N/A
5540705	12/1995	Meade et al.	N/A	N/A
5542949	12/1995	Yoon	N/A	N/A
5554185	12/1995	Block et al.	N/A	N/A
5562678	12/1995	Booker	N/A	N/A
5569274	12/1995	Rapacki et al.	N/A	N/A
5571085	12/1995	Accisano, III	N/A	N/A
5571137	12/1995	Marlow et al.	N/A	N/A
5571215	12/1995	Sterman et al.	N/A	N/A
5575802	12/1995	McQuilkin et al.	N/A	N/A
5582611	12/1995	Tsuruta et al.	N/A	N/A
5584803	12/1995	Stevens et al.	N/A	N/A
5593424	12/1996	Northrup III	N/A	N/A
5593435	12/1996	Carpentier et al.	N/A	N/A
5609598	12/1996	Laufer et al.	N/A	N/A
5617854	12/1996	Anand	N/A	N/A
5618306	12/1996	Roth et al.	N/A	N/A
5620452	12/1996	Yoon	N/A	N/A
5620461	12/1996	Muijs Van De Moer et al.	N/A	N/A
5626588	12/1996	Sauer et al.	N/A	N/A
5634932	12/1996	Schmidt	N/A	N/A
5636634	12/1996	Kordis et al.	N/A	N/A
5639277	12/1996	Mariant et al.	N/A	N/A
5640955	12/1996	Ockuly et al.	N/A	N/A
5649937	12/1996	Bito et al.	N/A	N/A
5662681	12/1996	Nash et al.	N/A	N/A
5669917	12/1996	Sauer et al.	N/A	N/A
5669919	12/1996	Sanders et al.	N/A	N/A
5690671	12/1996	McGurk et al.	N/A	N/A
5695504	12/1996	Gifford, III et al.	N/A	N/A
5695505	12/1996	Yoon	N/A	N/A
5702825	12/1996	Keita et al.	N/A	N/A
5706824	12/1997	Whittier	N/A	N/A
5709707	12/1997	Lock et al.	N/A	N/A
5713910	12/1997	Gordon et al.	N/A	N/A
5713911	12/1997	Racenet et al.	N/A	N/A
5715817	12/1997	Stevens-Wright et al.	N/A	N/A
5716367	12/1997	Koike et al.	N/A	N/A
5718725	12/1997	Sterman et al.	N/A	N/A
5719725	12/1997	Nakao	N/A	N/A
5722421	12/1997	Francese et al.	N/A	N/A
5725542	12/1997	Yoon	N/A	N/A
5725556	12/1997	Moser et al.	N/A	N/A
5738649	12/1997	Macoviak	N/A	N/A
5741271	12/1997	Nakao et al.	N/A	N/A
5741280	12/1997	Fleenor	N/A	N/A
5746747	12/1997	McKeating	N/A	N/A

5749828	12/1997	Solomon et al.	N/A	N/A
5759193	12/1997	Burbank et al.	N/A	N/A
5769812	12/1997	Stevens et al.	N/A	N/A
5769863	12/1997	Garrison	N/A	N/A
5772578	12/1997	Heimberger et al.	N/A	N/A
5782845	12/1997	Shewchuk	N/A	N/A
5797927	12/1997	Yoon	N/A	N/A
5797960	12/1997	Stevens et al.	N/A	N/A
5810847	12/1997	Laufer et al.	N/A	N/A
5810849	12/1997	Kontos	N/A	N/A
5810853	12/1997	Yoon	N/A	N/A
5810876	12/1997	Kelleher	N/A	N/A
5814029	12/1997	Hassett	N/A	N/A
5820591	12/1997	Thompson et al.	N/A	N/A
5820592	12/1997	Hammerslag	N/A	N/A
5820630	12/1997	Lind	N/A	N/A
5820631	12/1997	Nobles	N/A	N/A
5823955	12/1997	Kuck et al.	N/A	N/A
5823956	12/1997	Roth et al.	N/A	N/A
5824065	12/1997	Gross	N/A	N/A
5827237	12/1997	Macoviak et al.	N/A	N/A
5829447	12/1997	Stevens et al.	N/A	N/A
5833671	12/1997	Macoviak et al.	N/A	N/A
5836955	12/1997	Buelna et al.	N/A	N/A
5840081	12/1997	Andersen et al.	N/A	N/A
5843031	12/1997	Hermann et al.	N/A	N/A
5843103	12/1997	Wulfman	N/A	N/A
5849019	12/1997	Yoon	N/A	N/A
5853422	12/1997	Huebsch et al.	N/A	N/A
5855271	12/1998	Eubanks et al.	N/A	N/A
5855590	12/1998	Malecki et al.	N/A	N/A
5855614	12/1998	Stevens et al.	N/A	N/A
5860990	12/1998	Nobles et al.	N/A	N/A
5861003	12/1998	Latson et al.	N/A	N/A
5868733	12/1998	Ockuly et al.	N/A	N/A
5876399	12/1998	Chia et al.	N/A	N/A
5879307	12/1998	Chio et al.	N/A	N/A
5885271	12/1998	Hamilton et al.	N/A	N/A
5891160	12/1998	Williamson, IV et al.	N/A	N/A
5895404	12/1998	Ruiz	N/A	N/A
5895417	12/1998	Pomeranz et al.	N/A	N/A
5906620	12/1998	Nakao et al.	N/A	N/A
5908420	12/1998	Parins	N/A	N/A
5916147	12/1998	Boury	N/A	N/A
5928224	12/1998	Laufer	N/A	N/A
5944733	12/1998	Engelson	N/A	N/A
5947363	12/1998	Bolduc et al.	N/A	N/A
5954732	12/1998	Hart et al.	N/A	N/A
5957949	12/1998	Leonhard et al.	N/A	N/A

5957973	12/1998	Quiachon et al.	N/A	N/A
5972020	12/1998	Carpentier et al.	N/A	N/A
5972030	12/1998	Garrison et al.	N/A	N/A
5980455	12/1998	Daniel et al.	N/A	N/A
5989284	12/1998	Laufer	N/A	N/A
5997547	12/1998	Nakao et al.	N/A	N/A
6007546	12/1998	Snow et al.	N/A	N/A
6015417	12/1999	Reynolds, Jr.	N/A	N/A
6019722	12/1999	Spence et al.	N/A	N/A
6022360	12/1999	Reimels et al.	N/A	N/A
6033378	12/1999	Lundquist et al.	N/A	N/A
6033419	12/1999	Hamblin et al.	N/A	N/A
6036699	12/1999	Andreas et al.	N/A	N/A
6048351	12/1999	Gordon et al.	N/A	N/A
6053933	12/1999	Balazs et al.	N/A	N/A
6056769	12/1999	Epstein et al.	N/A	N/A
6059757	12/1999	Macoviak et al.	N/A	N/A
6060628	12/1999	Aoyama et al.	N/A	N/A
6060629	12/1999	Pham et al.	N/A	N/A
6063106	12/1999	Gibson	N/A	N/A
6066146	12/1999	Carroll et al.	N/A	N/A
6068628	12/1999	Fanton et al.	N/A	N/A
6068629	12/1999	Haissaguerre et al.	N/A	N/A
6077214	12/1999	Mortier et al.	N/A	N/A
6086600	12/1999	Kortenbach	N/A	N/A
6088889	12/1999	Luther et al.	N/A	N/A
6090118	12/1999	McGuckin, Jr.	N/A	N/A
6099505	12/1999	Ryan et al.	N/A	N/A
6099553	12/1999	Hart et al.	N/A	N/A
6110145	12/1999	Macoviak	N/A	N/A
6117144	12/1999	Nobles et al.	N/A	N/A
6117159	12/1999	Huebsch et al.	N/A	N/A
6123665	12/1999	Kawano	N/A	N/A
6123699	12/1999	Webster, Jr.	N/A	N/A
6126658	12/1999	Baker	N/A	N/A
6132447	12/1999	Dorsey	N/A	N/A
6136010	12/1999	Modesitt et al.	N/A	N/A
6139508	12/1999	Simpson	N/A	N/A
6143024	12/1999	Campbell et al.	N/A	N/A
6159240	12/1999	Sparer et al.	N/A	N/A
6162233	12/1999	Williamson, IV et al.	N/A	N/A
6165164	12/1999	Hill et al.	N/A	N/A
6165183	12/1999	Kuehn et al.	N/A	N/A
6165204	12/1999	Levinson et al.	N/A	N/A
6168614	12/2000	Andersen et al.	N/A	N/A
6171320	12/2000	Monassevitch	N/A	N/A
6174322	12/2000	Schneidt	N/A	N/A
6180059	12/2000	Divino, Jr. et al.	N/A	N/A
6182664	12/2000	Cosgrove	N/A	N/A

6187003	12/2000	Buyse et al.	N/A	N/A
6190408	12/2000	Melvin	N/A	N/A
6197043	12/2000	Davidson	N/A	N/A
6203531	12/2000	Ockuly et al.	N/A	N/A
6203553	12/2000	Robertson et al.	N/A	N/A
6206893	12/2000	Klein et al.	N/A	N/A
6206907	12/2000	Marino et al.	N/A	N/A
6210419	12/2000	Mayenberger et al.	N/A	N/A
6210432	12/2000	Solem et al.	N/A	N/A
6245079	12/2000	Nobles et al.	N/A	N/A
6264617	12/2000	Bales et al.	N/A	N/A
6267746	12/2000	Bumbalough	N/A	N/A
6267781	12/2000	Tu	N/A	N/A
6269819	12/2000	Oz et al.	N/A	N/A
6277555	12/2000	Duran et al.	N/A	N/A
6283127	12/2000	Sterman et al.	N/A	N/A
6283962	12/2000	Tu et al.	N/A	N/A
6299637	12/2000	Shaolian et al.	N/A	N/A
6306133	12/2000	Tu et al.	N/A	N/A
6312447	12/2000	Grimes	N/A	N/A
6319250	12/2000	Falwell et al.	N/A	N/A
6322559	12/2000	Daulton et al.	N/A	N/A
6332893	12/2000	Mortier et al.	N/A	N/A
6334860	12/2001	Dorn	N/A	N/A
6352708	12/2001	Duran et al.	N/A	N/A
6355030	12/2001	Aldrich et al.	N/A	N/A
6358277	12/2001	Duran	N/A	N/A
6368326	12/2001	Dakin et al.	N/A	N/A
6387104	12/2001	Pugsley, Jr. et al.	N/A	N/A
6402780	12/2001	Williamson et al.	N/A	N/A
6402781	12/2001	Langberg et al.	N/A	N/A
6406420	12/2001	McCarthy et al.	N/A	N/A
6419640	12/2001	Taylor	N/A	N/A
6419669	12/2001	Frazier et al.	N/A	N/A
6461366	12/2001	Seguin	N/A	N/A
6464707	12/2001	Bjerken	N/A	N/A
6482224	12/2001	Michler et al.	N/A	N/A
6485489	12/2001	Teirstein et al.	N/A	N/A
6494881	12/2001	Bales et al.	N/A	N/A
6508828	12/2002	Akerfeldt et al.	N/A	N/A
6517550	12/2002	Konya et al.	N/A	N/A
6533796	12/2002	Sauer et al.	N/A	N/A
6537314	12/2002	Langberg et al.	N/A	N/A
6540755	12/2002	Ockuly et al.	N/A	N/A
6551331	12/2002	Nobles et al.	N/A	N/A
6562037	12/2002	Paton et al.	N/A	N/A
6562052	12/2002	Nobles et al.	N/A	N/A
6575971	12/2002	Hauck et al.	N/A	N/A
6585761	12/2002	Taheri	N/A	N/A
6599311	12/2002	Biggs et al.	N/A	N/A

6616684	12/2002	Vidlund et al.	N/A	N/A
6619291	12/2002	Hlavka et al.	N/A	N/A
6626899	12/2002	Houser et al.	N/A	N/A
6626921	12/2002	Blatter et al.	N/A	N/A
6626930	12/2002	Allen et al.	N/A	N/A
6629534	12/2002	St. Goar et al.	N/A	N/A
6641592	12/2002	Sauer et al.	N/A	N/A
6656221	12/2002	Taylor et al.	N/A	N/A
6669687	12/2002	Saadat	N/A	N/A
6685648	12/2003	Flaherty et al.	N/A	N/A
6689164	12/2003	Seguin	N/A	N/A
6695866	12/2003	Kuehn et al.	N/A	N/A
6701929	12/2003	Hussein	N/A	N/A
6702825	12/2003	Frazier et al.	N/A	N/A
6702826	12/2003	Liddicoat et al.	N/A	N/A
6709382	12/2003	Homer	N/A	N/A
6709456	12/2003	Langberg et al.	N/A	N/A
6718985	12/2003	Hlavka et al.	N/A	N/A
6719767	12/2003	Kimblad	N/A	N/A
6723038	12/2003	Schroeder et al.	N/A	N/A
6726716	12/2003	Marquez	N/A	N/A
6740107	12/2003	Loeb et al.	N/A	N/A
6746471	12/2003	Mortier et al.	N/A	N/A
6752813	12/2003	Goldfarb et al.	N/A	N/A
6755777	12/2003	Schweich et al.	N/A	N/A
6764510	12/2003	Vidlund et al.	N/A	N/A
6767349	12/2003	Ouchi	N/A	N/A
6770083	12/2003	Seguin	N/A	N/A
6797001	12/2003	Mathis et al.	N/A	N/A
6797002	12/2003	Spence et al.	N/A	N/A
6860179	12/2004	Hopper et al.	N/A	N/A
6875224	12/2004	Grimes	N/A	N/A
6926715	12/2004	Hauck et al.	N/A	N/A
6932810	12/2004	Ryan	N/A	N/A
6945978	12/2004	Hyde	N/A	N/A
6949122	12/2004	Adams et al.	N/A	N/A
6966914	12/2004	Abe	N/A	N/A
6986775	12/2005	Morales et al.	N/A	N/A
7004970	12/2005	Cauthen III et al.	N/A	N/A
7011669	12/2005	Kimblad	N/A	N/A
7033390	12/2005	Johnson et al.	N/A	N/A
7048754	12/2005	Martin et al.	N/A	N/A
7056294	12/2005	Khairkhahan et al.	N/A	N/A
7112207	12/2005	Allen et al.	N/A	N/A
7226467	12/2006	Lucatero et al.	N/A	N/A
7258694	12/2006	Choi et al.	N/A	N/A
7288097	12/2006	Seguin	N/A	N/A
7291168	12/2006	Macoviak et al.	N/A	N/A
7338467	12/2007	Lutter	N/A	N/A
7381210	12/2007	Zarbatany et al.	N/A	N/A

7435257	12/2007	Lashinski et al.	N/A	N/A
7464712	12/2007	Oz et al.	N/A	N/A
7497822	12/2008	Kugler et al.	N/A	N/A
7533790	12/2008	Knodel et al.	N/A	N/A
7563267	12/2008	Goldfarb et al.	N/A	N/A
7563273	12/2008	Goldfarb et al.	N/A	N/A
7604646	12/2008	Goldfarb et al.	N/A	N/A
7608091	12/2008	Goldfarb et al.	N/A	N/A
7635329	12/2008	Goldfarb et al.	N/A	N/A
7651502	12/2009	Jackson	N/A	N/A
7655015	12/2009	Goldfarb et al.	N/A	N/A
7666204	12/2009	Thornton et al.	N/A	N/A
7955340	12/2010	Michlitsch	N/A	N/A
8216234	12/2011	Long	N/A	N/A
8257356	12/2011	Bleich et al.	N/A	N/A
8398708	12/2012	Meiri et al.	N/A	N/A
8435237	12/2012	Bahney	N/A	N/A
8496655	12/2012	Epp	606/45	A61B 18/14
8500768	12/2012	Cohen	N/A	N/A
8523881	12/2012	Cabiri et al.	N/A	N/A
8623077	12/2013	Cohn	N/A	N/A
8690858	12/2013	Machold et al.	N/A	N/A
8821518	12/2013	Saliman et al.	N/A	N/A
8926588	12/2014	Berthiaume et al.	N/A	N/A
9126032	12/2014	Khairkhahan et al.	N/A	N/A
9211119	12/2014	Hendricksen	N/A	N/A
9370341	12/2015	Ceniccola et al.	N/A	N/A
9498331	12/2015	Chang et al.	N/A	N/A
9572666	12/2016	Basude et al.	N/A	N/A
9770256	12/2016	Cohen et al.	N/A	N/A
9949833	12/2017	McCleary et al.	N/A	N/A
10238493	12/2018	Metchik et al.	N/A	N/A
10667804	12/2019	Basude et al.	N/A	N/A
11013554	12/2020	Coates	N/A	N/A
11406250	12/2021	Saadat et al.	N/A	N/A
2001/0002445	12/2000	Vesely	N/A	N/A
2001/0004715	12/2000	Duran et al.	N/A	N/A
2001/0005787	12/2000	Oz et al.	N/A	N/A
2001/0010005	12/2000	Kammerer et al.	N/A	N/A
2001/0018611	12/2000	Solem et al.	N/A	N/A
2001/0022872	12/2000	Marui	N/A	N/A
2001/0037084	12/2000	Nardeo	N/A	N/A
2001/0039411	12/2000	Johansson et al.	N/A	N/A
2001/0044568	12/2000	Langberg et al.	N/A	N/A
2001/0044635	12/2000	Niizeki et al.	N/A	N/A
2002/0013547	12/2001	Paskar	N/A	N/A
2002/0013571	12/2001	Goldfarb et al.	N/A	N/A
2002/0022848	12/2001	Garrison et al.	N/A	N/A
2002/0026233	12/2001	Shaknovich	N/A	N/A
2002/0035361	12/2001	Houser et al.	N/A	N/A

2002/0035381	12/2001	Bardy et al.	N/A	N/A
2002/0042651	12/2001	Liddicoat et al.	N/A	N/A
2002/0055767	12/2001	Forde et al.	N/A	N/A
2002/0055774	12/2001	Liddicoat	N/A	N/A
2002/0055775	12/2001	Carpentier et al.	N/A	N/A
2002/0058910	12/2001	Hermann et al.	N/A	N/A
2002/0058995	12/2001	Stevens	N/A	N/A
2002/0077687	12/2001	Ahn	N/A	N/A
2002/0087148	12/2001	Brock et al.	N/A	N/A
2002/0087169	12/2001	Brock et al.	N/A	N/A
2002/0087173	12/2001	Alferness et al.	N/A	N/A
2002/0103532	12/2001	Langberg et al.	N/A	N/A
2002/0107534	12/2001	Schaefer et al.	N/A	N/A
2002/0147456	12/2001	Diduch et al.	N/A	N/A
2002/0156526	12/2001	Hilavka et al.	N/A	N/A
2002/0158528	12/2001	Tsuzaki et al.	N/A	N/A
2002/0161378	12/2001	Downing	N/A	N/A
2002/0169360	12/2001	Taylor et al.	N/A	N/A
2002/0173811	12/2001	Tu et al.	N/A	N/A
2002/0173841	12/2001	Ortiz et al.	N/A	N/A
2002/0183766	12/2001	Seguin	N/A	N/A
2002/0183787	12/2001	Wahr et al.	N/A	N/A
2002/0183835	12/2001	Taylor et al.	N/A	N/A
2003/0005797	12/2002	Hopper et al.	N/A	N/A
2003/0045778	12/2002	Ohline et al.	N/A	N/A
2003/0050693	12/2002	Quijano et al.	N/A	N/A
2003/0069570	12/2002	Witzel et al.	N/A	N/A
2003/0069593	12/2002	Tremulis et al.	N/A	N/A
2003/0069636	12/2002	Solem et al.	N/A	N/A
2003/0074012	12/2002	Nguyen et al.	N/A	N/A
2003/0078654	12/2002	Taylor et al.	N/A	N/A
2003/0083742	12/2002	Spence et al.	N/A	N/A
2003/0105519	12/2002	Fasol et al.	N/A	N/A
2003/0105520	12/2002	Alferness et al.	N/A	N/A
2003/0120340	12/2002	Lisk et al.	N/A	N/A
2003/0120341	12/2002	Shennib et al.	N/A	N/A
2003/0130669	12/2002	Damarati	N/A	N/A
2003/0130730	12/2002	Cohn et al.	N/A	N/A
2003/0144697	12/2002	Mathis et al.	N/A	N/A
2003/0167071	12/2002	Martin et al.	N/A	N/A
2003/0171776	12/2002	Adams et al.	N/A	N/A
2003/0187467	12/2002	Schreck	N/A	N/A
2003/0195562	12/2002	Collier et al.	N/A	N/A
2003/0208231	12/2002	Williamson, IV et al.	N/A	N/A
2003/0229395	12/2002	Cox	N/A	N/A
2003/0233038	12/2002	Hassett	N/A	N/A
2004/0002719	12/2003	Oz et al.	N/A	N/A
2004/0003819	12/2003	St. Goar et al.	N/A	N/A
2004/0015232	12/2003	Shu et al.	N/A	N/A

2004/0019377	12/2003	Taylor et al.	N/A	N/A
2004/0019378	12/2003	Hlavka et al.	N/A	N/A
2004/0024414	12/2003	Downing	N/A	N/A
2004/0030319	12/2003	Korkor et al.	N/A	N/A
2004/0030382	12/2003	St. Goar et al.	N/A	N/A
2004/0034380	12/2003	Woolfson et al.	N/A	N/A
2004/0039442	12/2003	St. Goar et al.	N/A	N/A
2004/0039443	12/2003	Solem et al.	N/A	N/A
2004/0044350	12/2003	Martin et al.	N/A	N/A
2004/0044365	12/2003	Bachman	N/A	N/A
2004/0049207	12/2003	Goldfarb et al.	N/A	N/A
2004/0049211	12/2003	Tremulis et al.	N/A	N/A
2004/0059345	12/2003	Nakao et al.	N/A	N/A
2004/0073302	12/2003	Rourke et al.	N/A	N/A
2004/0078053	12/2003	Berg et al.	N/A	N/A
2004/0087975	12/2003	Lucatero et al.	N/A	N/A
2004/0088047	12/2003	Spence et al.	N/A	N/A
2004/0092858	12/2003	Wilson et al.	N/A	N/A
2004/0092962	12/2003	Thornton et al.	N/A	N/A
2004/0097878	12/2003	Anderson et al.	N/A	N/A
2004/0097979	12/2003	Svanidze et al.	N/A	N/A
2004/0106989	12/2003	Wilson et al.	N/A	N/A
2004/0111099	12/2003	Nguyen et al.	N/A	N/A
2004/0116848	12/2003	Gardeski et al.	N/A	N/A
2004/0116951	12/2003	Rosengart	N/A	N/A
2004/0122448	12/2003	Levine	N/A	N/A
2004/0127849	12/2003	Kantor	N/A	N/A
2004/0127981	12/2003	Randert et al.	N/A	N/A
2004/0127982	12/2003	Machold et al.	N/A	N/A
2004/0127983	12/2003	Mortier et al.	N/A	N/A
2004/0133062	12/2003	Pai et al.	N/A	N/A
2004/0133063	12/2003	McCarthy et al.	N/A	N/A
2004/0133082	12/2003	Abraham-Fuchs et al.	N/A	N/A
2004/0133192	12/2003	Houser et al.	N/A	N/A
2004/0133220	12/2003	Lashinski et al.	N/A	N/A
2004/0133232	12/2003	Rosenbluth et al.	N/A	N/A
2004/0133240	12/2003	Adams et al.	N/A	N/A
2004/0133273	12/2003	Cox	N/A	N/A
2004/0138744	12/2003	Lashinski et al.	N/A	N/A
2004/0138745	12/2003	Macoviak et al.	N/A	N/A
2004/0147826	12/2003	Peterson	N/A	N/A
2004/0148021	12/2003	Cartledge et al.	N/A	N/A
2004/0152847	12/2003	Emri et al.	N/A	N/A
2004/0152947	12/2003	Schroeder et al.	N/A	N/A
2004/0153144	12/2003	Seguin	N/A	N/A
2004/0158123	12/2003	Jayaraman	N/A	N/A
2004/0162610	12/2003	Laiska et al.	N/A	N/A
2004/0167539	12/2003	Kuehn et al.	N/A	N/A
2004/0186486	12/2003	Roue et al.	N/A	N/A

2004/0186566	12/2003	Hindrichs et al.	N/A	N/A
2004/0193191	12/2003	Starksen et al.	N/A	N/A
2004/0215339	12/2003	Drasler et al.	N/A	N/A
2004/0220593	12/2003	Greenhalgh	N/A	N/A
2004/0220657	12/2003	Nieminen et al.	N/A	N/A
2004/0225233	12/2003	Frankowski et al.	N/A	N/A
2004/0225300	12/2003	Goldfarb et al.	N/A	N/A
2004/0225305	12/2003	Ewers et al.	N/A	N/A
2004/0225353	12/2003	McGuckin, Jr.	623/2.11	A61F 2/2421
2004/0236354	12/2003	Seguin	N/A	N/A
2004/0242960	12/2003	Orban	N/A	N/A
2004/0243229	12/2003	Vidlund et al.	N/A	N/A
2004/0249452	12/2003	Adams et al.	N/A	N/A
2004/0249453	12/2003	Cartledge et al.	N/A	N/A
2004/0260393	12/2003	Randert et al.	N/A	N/A
2005/0004583	12/2004	Oz et al.	N/A	N/A
2005/0004665	12/2004	Aklog	N/A	N/A
2005/0004668	12/2004	Aklog et al.	N/A	N/A
2005/0021056	12/2004	St. Goer et al.	N/A	N/A
2005/0021057	12/2004	St. Goer et al.	N/A	N/A
2005/0021058	12/2004	Negro	N/A	N/A
2005/0033446	12/2004	Deem et al.	N/A	N/A
2005/0038383	12/2004	Kelley et al.	N/A	N/A
2005/0038508	12/2004	Gabbay	N/A	N/A
2005/0049698	12/2004	Bolling et al.	N/A	N/A
2005/0055089	12/2004	Macoviak et al.	N/A	N/A
2005/0059351	12/2004	Cauwels et al.	N/A	N/A
2005/0065453	12/2004	Shabaz et al.	N/A	N/A
2005/0085903	12/2004	Lau	N/A	N/A
2005/0119735	12/2004	Spence et al.	N/A	N/A
2005/0131438	12/2004	Cohn	N/A	N/A
2005/0143809	12/2004	Salahieh et al.	N/A	N/A
2005/0149014	12/2004	Hauck et al.	N/A	N/A
2005/0159763	12/2004	Mollenauer	N/A	N/A
2005/0159810	12/2004	Filsoufi	N/A	N/A
2005/0192633	12/2004	Montpetit	N/A	N/A
2005/0197694	12/2004	Pai et al.	N/A	N/A
2005/0197695	12/2004	Stacchino et al.	N/A	N/A
2005/0216039	12/2004	Lederman	N/A	N/A
2005/0228422	12/2004	Machold et al.	N/A	N/A
2005/0228495	12/2004	Macoviak	N/A	N/A
2005/0251001	12/2004	Hassett	N/A	N/A
2005/0256452	12/2004	DeMarchi et al.	N/A	N/A
2005/0267493	12/2004	Schreck et al.	N/A	N/A
2005/0273160	12/2004	Lashinski et al.	N/A	N/A
2005/0277876	12/2004	Hayden	N/A	N/A
2005/0287493	12/2004	Novak et al.	N/A	N/A
2006/0004247	12/2005	Kute et al.	N/A	N/A
2006/0009759	12/2005	Chrisitian et al.	N/A	N/A

2006/0015003	12/2005	Moaddes et al.	N/A	N/A
2006/0015179	12/2005	Bulman-Fleming et al.	N/A	N/A
2006/0020275	12/2005	Goldfarb et al.	N/A	N/A
2006/0020327	12/2005	Lashinski et al.	N/A	N/A
2006/0020334	12/2005	Lashinski et al.	N/A	N/A
2006/0030866	12/2005	Schreck	N/A	N/A
2006/0030867	12/2005	Zadno	N/A	N/A
2006/0030885	12/2005	Hyde	N/A	N/A
2006/0058871	12/2005	Zakay et al.	N/A	N/A
2006/0064115	12/2005	Allen et al.	N/A	N/A
2006/0064116	12/2005	Allen et al.	N/A	N/A
2006/0064118	12/2005	Kimblad	N/A	N/A
2006/0074484	12/2005	Huber	N/A	N/A
2006/0089671	12/2005	Goldfarb et al.	N/A	N/A
2006/0089711	12/2005	Dolan	N/A	N/A
2006/0135961	12/2005	Rosenman et al.	N/A	N/A
2006/0135993	12/2005	Seguin	N/A	N/A
2006/0184198	12/2005	Bales	N/A	N/A
2006/0184203	12/2005	Martin et al.	N/A	N/A
2006/0195012	12/2005	Mortier et al.	N/A	N/A
2006/0229708	12/2005	Powell et al.	N/A	N/A
2006/0252984	12/2005	Randert et al.	N/A	N/A
2006/0276890	12/2005	Solem et al.	N/A	N/A
2007/0016225	12/2006	Nakao	N/A	N/A
2007/0038293	12/2006	St. Goar et al.	N/A	N/A
2007/0060997	12/2006	de Boer	N/A	N/A
2007/0073185	12/2006	Nakao	N/A	N/A
2007/0100356	12/2006	Lucatero et al.	N/A	N/A
2007/0118155	12/2006	Goldfarb et al.	N/A	N/A
2007/0129737	12/2006	Goldfarb et al.	N/A	N/A
2007/0173757	12/2006	Levine et al.	N/A	N/A
2007/0197858	12/2006	Goldfarb et al.	N/A	N/A
2007/0198082	12/2006	Kapadia et al.	N/A	N/A
2007/0260225	12/2006	Sakakine et al.	N/A	N/A
2007/0287884	12/2006	Schena	N/A	N/A
2008/0009858	12/2007	Rizvi	N/A	N/A
2008/0039935	12/2007	Buch et al.	N/A	N/A
2008/0045936	12/2007	Vaska et al.	N/A	N/A
2008/0051703	12/2007	Thorton et al.	N/A	N/A
2008/0051807	12/2007	St. Goar et al.	N/A	N/A
2008/0097467	12/2007	Gruber et al.	N/A	N/A
2008/0097489	12/2007	Goldfarb et al.	N/A	N/A
2008/0167714	12/2007	St. Goer et al.	N/A	N/A
2008/0183194	12/2007	Goldfarb et al.	N/A	N/A
2008/0188850	12/2007	Mody et al.	N/A	N/A
2008/0195126	12/2007	Solem	N/A	N/A
2008/0243249	12/2007	Kohm et al.	N/A	N/A
2008/0294175	12/2007	Bardsley et al.	N/A	N/A
2008/0312496	12/2007	Zwolinski	N/A	N/A

2009/0012538	12/2008	Saliman et al.	N/A	N/A
2009/0036768	12/2008	Seehusen et al.	N/A	N/A
2009/0156995	12/2008	Martin et al.	N/A	N/A
2009/0163934	12/2008	Raschdorf, Jr. et al.	N/A	N/A
2009/0177266	12/2008	Powell et al.	N/A	N/A
2009/0192510	12/2008	Bahney	N/A	N/A
2009/0198322	12/2008	Deem et al.	N/A	N/A
2009/0204005	12/2008	Keast et al.	N/A	N/A
2009/0209955	12/2008	Forster et al.	N/A	N/A
2009/0209991	12/2008	Hinchliffe et al.	N/A	N/A
2009/0270858	12/2008	Hauck et al.	N/A	N/A
2009/0276039	12/2008	Meretei	N/A	N/A
2009/0281619	12/2008	Le et al.	N/A	N/A
2009/0326567	12/2008	Goldfarb et al.	N/A	N/A
2010/0016958	12/2009	St. Goer et al.	N/A	N/A
2010/0022823	12/2009	Goldfarb et al.	N/A	N/A
2010/0044410	12/2009	Argentine et al.	N/A	N/A
2010/0121437	12/2009	Subramanian et al.	N/A	N/A
2010/0152612	12/2009	Headley et al.	N/A	N/A
2010/0217261	12/2009	Watson	N/A	N/A
2010/0262231	12/2009	Tuval et al.	N/A	N/A
2010/0268226	12/2009	Epp et al.	N/A	N/A
2010/0298929	12/2009	Thornton et al.	N/A	N/A
2011/0009864	12/2010	Bucciaglia et al.	N/A	N/A
2011/0184405	12/2010	Mueller	N/A	N/A
2011/0224710	12/2010	Bleich	N/A	N/A
2011/0238052	12/2010	Robinson	N/A	N/A
2011/0282250	12/2010	Fung et al.	N/A	N/A
2012/0022527	12/2011	Woodruff	N/A	N/A
2012/0022640	12/2011	Gross et al.	N/A	N/A
2012/0065464	12/2011	Ellis et al.	N/A	N/A
2012/0150194	12/2011	Odermatt et al.	N/A	N/A
2012/0157765	12/2011	Mitelberg	N/A	N/A
2012/0172915	12/2011	Fifer et al.	N/A	N/A
2012/0179184	12/2011	Orlov	N/A	N/A
2012/0265222	12/2011	Gordin et al.	N/A	N/A
2012/0310330	12/2011	Buchbinder et al.	N/A	N/A
2012/0316639	12/2011	Kleinschrodt	N/A	N/A
2012/0330348	12/2011	Strauss et al.	N/A	N/A
2013/0041314	12/2012	Dillon	N/A	N/A
2013/0066341	12/2012	Ketai et al.	N/A	N/A
2013/0066342	12/2012	Dell et al.	N/A	N/A
2013/0109910	12/2012	Alexander et al.	N/A	N/A
2013/0172828	12/2012	Kappel et al.	N/A	N/A
2013/0317515	12/2012	Kuroda et al.	N/A	N/A
2014/0039511	12/2013	Morris et al.	N/A	N/A
2014/0135799	12/2013	Henderson	N/A	N/A
2014/0228871	12/2013	Cohen et al.	N/A	N/A
2014/0276913	12/2013	Tah et al.	N/A	N/A
2014/0309670	12/2013	Bakos et al.	N/A	N/A

2014/0324164	12/2013	Gross et al.	N/A	N/A
2014/0350662	12/2013	Vaturi	N/A	N/A
2014/0358224	12/2013	Tegels et al.	N/A	N/A
2014/0364866	12/2013	Dryden et al.	N/A	N/A
2014/0379074	12/2013	Spence et al.	N/A	N/A
2015/0005704	12/2014	Heisel et al.	N/A	N/A
2015/0005801	12/2014	Marquis et al.	N/A	N/A
2015/0051698	12/2014	Ruyra Baliarda et al.	N/A	N/A
2015/0094800	12/2014	Chawla	N/A	N/A
2015/0112430	12/2014	Creaven et al.	N/A	N/A
2015/0211946	12/2014	Pons et al.	N/A	N/A
2015/0230947	12/2014	Krieger et al.	N/A	N/A
2015/0257877	12/2014	Hernandez	N/A	N/A
2015/0257883	12/2014	Basude et al.	N/A	N/A
2015/0306806	12/2014	Dando et al.	N/A	N/A
2015/0313581	12/2014	Wolfe et al.	N/A	N/A
2016/0015410	12/2015	Asirvatham et al.	N/A	N/A
2016/0074165	12/2015	Spence et al.	N/A	N/A
2016/0174979	12/2015	Wei	N/A	N/A
2016/0317174	12/2015	Dake	N/A	N/A
2017/0042678	12/2016	Ganesan et al.	N/A	N/A
2017/0100183	12/2016	Iaizzo et al.	N/A	N/A
2017/0143330	12/2016	Basude et al.	N/A	N/A
2017/0202559	12/2016	Taha	N/A	N/A
2017/0232238	12/2016	Biller et al.	N/A	N/A
2018/0008268	12/2017	Khairkhahan	N/A	N/A
2018/0028215	12/2017	Cohen	N/A	N/A
2018/0092661	12/2017	Prabhu	N/A	N/A
2018/0133010	12/2017	Kizuka	N/A	N/A
2018/0161159	12/2017	Lee	N/A	N/A
2018/0360457	12/2017	Ellis et al.	N/A	N/A
2019/0029790	12/2018	Bak-Boychuk et al.	N/A	N/A
2019/0183571	12/2018	De Marchena	N/A	N/A
2019/0298517	12/2018	Sanchez et al.	N/A	N/A
2019/0307458	12/2018	Mathis et al.	N/A	N/A
2020/0121460	12/2019	Dale et al.	N/A	N/A
2021/0113232	12/2020	Ortiz et al.	N/A	N/A
2021/0145574	12/2020	Childs et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1469724	12/2003	CN	N/A
102770080	12/2011	CN	N/A
103841899	12/2013	CN	N/A
104244841	12/2013	CN	N/A
3504292	12/1985	DE	N/A
9100873	12/1990	DE	N/A

10116168	12/2000	DE	N/A
0179562	12/1988	EP	N/A
0558031	12/1992	EP	N/A
0684012	12/1994	EP	N/A
0727239	12/1995	EP	N/A
0782836	12/1996	EP	N/A
1230899	12/2001	EP	N/A
1674040	12/2005	EP	N/A
1980288	12/2007	EP	N/A
2005912	12/2007	EP	N/A
2537487	12/2011	EP	N/A
2641570	12/2012	EP	N/A
2702965	12/2013	EP	N/A
2740419	12/2013	EP	N/A
3009103	12/2015	EP	N/A
2705556	12/1993	FR	N/A
2768324	12/1998	FR	N/A
2903292	12/2007	FR	N/A
1598111	12/1980	GB	N/A
2151142	12/1984	GB	N/A
09-253030	12/1996	JP	N/A
11-089937	12/1998	JP	N/A
2000-283130	12/1999	JP	N/A
2001-517529	12/2000	JP	N/A
2006528911	12/2005	JP	N/A
2013516244	12/2012	JP	N/A
2013-523384	12/2012	JP	N/A
2014523274	12/2013	JP	N/A
2015502548	12/2014	JP	N/A
2018-030008	12/2017	JP	N/A
WO 1981000668	12/1980	WO	N/A
WO 1991001689	12/1990	WO	N/A
91/18881	12/1990	WO	N/A
92/12690	12/1991	WO	N/A
WO 1994018881	12/1993	WO	N/A
WO 1994018893	12/1993	WO	N/A
WO 1995008292	12/1994	WO	N/A
WO 1995011620	12/1994	WO	N/A
WO 1995015715	12/1994	WO	N/A
96/14032	12/1995	WO	N/A
WO 1996020655	12/1995	WO	N/A
WO 1996022735	12/1995	WO	N/A
WO 1996030072	12/1995	WO	N/A
WO 1997018746	12/1996	WO	N/A
WO 1997025927	12/1996	WO	N/A
WO 1997026034	12/1996	WO	N/A
WO 1997038748	12/1996	WO	N/A
WO 1997039688	12/1996	WO	N/A
WO 1997048436	12/1996	WO	N/A
WO 1998007375	12/1997	WO	N/A

WO 1998024372	12/1997	WO	N/A
WO 1998030153	12/1997	WO	N/A
WO 1998032382	12/1997	WO	N/A
WO 1998035638	12/1997	WO	N/A
WO 1999000059	12/1998	WO	N/A
WO 1999001377	12/1998	WO	N/A
99/07354	12/1998	WO	N/A
WO 1999007295	12/1998	WO	N/A
WO 1999013777	12/1998	WO	N/A
WO 1999044524	12/1998	WO	N/A
99/66967	12/1998	WO	N/A
00/02489	12/1999	WO	N/A
WO 2000003651	12/1999	WO	N/A
WO 2000003759	12/1999	WO	N/A
WO 2000012168	12/1999	WO	N/A
WO 2000044313	12/1999	WO	N/A
00/59382	12/1999	WO	N/A
WO 2000060995	12/1999	WO	N/A
01/00111	12/2000	WO	N/A
01/03651	12/2000	WO	N/A
WO 2001000114	12/2000	WO	N/A
01/26586	12/2000	WO	N/A
01/26587	12/2000	WO	N/A
01/26588	12/2000	WO	N/A
01/26703	12/2000	WO	N/A
WO 2001026557	12/2000	WO	N/A
WO 2001028432	12/2000	WO	N/A
WO 2001028455	12/2000	WO	N/A
WO 2001047438	12/2000	WO	N/A
WO 2001049213	12/2000	WO	N/A
WO 2001050985	12/2000	WO	N/A
01/56512	12/2000	WO	N/A
WO 2001054618	12/2000	WO	N/A
WO 2001066001	12/2000	WO	N/A
WO 2001070320	12/2000	WO	N/A
WO 2001089440	12/2000	WO	N/A
01/95831	12/2000	WO	N/A
01/95832	12/2000	WO	N/A
01/97741	12/2000	WO	N/A
WO 2002000099	12/2001	WO	N/A
WO 2002001999	12/2001	WO	N/A
WO 2002003892	12/2001	WO	N/A
WO 2002034167	12/2001	WO	N/A
02/62270	12/2001	WO	N/A
02/62408	12/2001	WO	N/A
WO 2002060352	12/2001	WO	N/A
WO 2002062263	12/2001	WO	N/A
WO 2003001893	12/2002	WO	N/A
WO 2003003930	12/2002	WO	N/A
WO 2003020179	12/2002	WO	N/A

WO 2003028558	12/2002	WO	N/A
WO 2003037171	12/2002	WO	N/A
WO 2003047467	12/2002	WO	N/A
WO 2003049619	12/2002	WO	N/A
WO 2003073910	12/2002	WO	N/A
WO 2003073913	12/2002	WO	N/A
03/82129	12/2002	WO	N/A
WO 2003105667	12/2002	WO	N/A
2004/004607	12/2003	WO	N/A
WO 2004006810	12/2003	WO	N/A
WO 2004012583	12/2003	WO	N/A
WO 2004012789	12/2003	WO	N/A
WO 2004014282	12/2003	WO	N/A
WO 2004019811	12/2003	WO	N/A
WO 2004030570	12/2003	WO	N/A
WO 2004037317	12/2003	WO	N/A
2004/047679	12/2003	WO	N/A
WO 2004045370	12/2003	WO	N/A
WO 2004045378	12/2003	WO	N/A
WO 2004045463	12/2003	WO	N/A
WO 2004062725	12/2003	WO	N/A
WO 2004082523	12/2003	WO	N/A
WO 2004082538	12/2003	WO	N/A
2004/093730	12/2003	WO	N/A
2004/112651	12/2003	WO	N/A
WO 2004103162	12/2003	WO	N/A
WO 2004112585	12/2003	WO	N/A
2005/002424	12/2004	WO	N/A
2005/027797	12/2004	WO	N/A
WO 2005018507	12/2004	WO	N/A
WO 2005032421	12/2004	WO	N/A
WO 2005062931	12/2004	WO	N/A
WO 2005112792	12/2004	WO	N/A
WO 2006037073	12/2005	WO	N/A
WO 2006105008	12/2005	WO	N/A
WO 2006105009	12/2005	WO	N/A
WO 2006113906	12/2005	WO	N/A
2006/115875	12/2005	WO	N/A
WO 2006115876	12/2005	WO	N/A
WO 2007136829	12/2006	WO	N/A
WO 2008103722	12/2007	WO	N/A
WO 2010024801	12/2009	WO	N/A
WO 2010121076	12/2009	WO	N/A
WO 2012020521	12/2011	WO	N/A
WO 2013049734	12/2012	WO	N/A
WO 2013103934	12/2012	WO	N/A
WO 2014064694	12/2013	WO	N/A
WO 2014121280	12/2013	WO	N/A
WO 2016022797	12/2015	WO	N/A
WO 2016144708	12/2015	WO	N/A

WO 2016150806	12/2015	WO	N/A
WO 2017223073	12/2016	WO	N/A
WO 2018009718	12/2017	WO	N/A
WO 2018106482	12/2017	WO	N/A
2018/236766	12/2017	WO	N/A
2019/040943	12/2018	WO	N/A
2019/195336	12/2018	WO	N/A

OTHER PUBLICATIONS

Nishimura, et al. 2014 AHA/ACC guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol. Jun. 10, 2014; 63(22):2438-88. cited by applicant

Abe et al, "De Vega's Annuloplasty for Acquired Tricuspid Disease: Early and Late Results in 110 Patients", Ann. Thorac. Surg., pp. 670-676, vol. 48 (Jan. 1989). cited by applicant

Abe et al., "Updated in 1996—De Vega's Annuloplasty for Acquired Tricuspid Disease: Early and Late Results in 110 Patients", Ann. Thorac. Surg., pp. 1876-1877, vol. 62 (1996). cited by applicant

Alfieri et al., "An Effective Technique to Correct Anterior Mitral Leaflet Prolapse," J. Card Surg., 14:468-470 (1999). cited by applicant

Alfieri et al., "Novel Suture Device for Beating Heart Mitral Leaflet Approximation," Annals of Thoracic Surgery, 74:1488-1493 (2002). cited by applicant

Alfieri et al., "The double orifice technique in mitral valve repair: a simple solution for complex problems," Journal of Thoracic and Cardiovascular Surgery, 122:674-681 (2001). cited by applicant

Alfieri et al., "The edge to edge technique," The European Association for Cardio-Thoracic Surgery 14th Annual Meeting, Oct. 7-11, 2000, Book of Proceedings. cited by applicant

Ali Khan et al, Blade Atrial Septostomy: Experience with the First 50 Procedures, Cathet. Cardiovasc. Diagn., Aug. 1991, pp. 257-262, vol. 23. cited by applicant

Alvarez et al, Repairing the Degenerative Mitral Valve: Ten to Fifteen-year Follow-up, Journal of Thoracic Cardiovascular Surgery, Aug. 1996, pp. 238-247, vol. 112, No. 2. cited by applicant

Bailey, "Mitral Regurgitation" in Surgery of the Heart, Chapter 20, pp. 686-737 (1955). cited by applicant

Borghetti et al., "Preliminary observations on haemodynamics during physiological stress conditions following 'double-orifice' mitral valve repair," European Journal of Cardio-thoracic Surgery, 20:262-269 (2001). cited by applicant

Castedo, "Edge-to-Edge Tricuspid Repair for Redeveloped Valve Incompetence after DeVega's Annuloplasty," Ann Thora Surg., 75:605-606 (2003). cited by applicant

Chinese Office Action issued in Chinese Application No. 200980158707.2 dated Sep. 9, 2013. cited by applicant

Communication dated Apr. 16, 2018 from the European Patent Office in counterpart European application No. 04752603.3. cited by applicant

Communication dated Apr. 28, 2017 issued by the European Patent Office in counterpart application No. 16196023.2. cited by applicant

Communication dated Jan. 26, 2017, from the European Patent Office in counterpart European application No. 16196023.2. cited by applicant

Dang N C et al., "Surgical Revision After Percutaneous Mitral Valve Repair with a Clip: Initial Multicenter Experience", The Annals of Thoracic Surgery, Elsevier, United States, vol. 80, No. 6, pp. 2338-2342, (Dec. 1, 2005), XP027732951, ISSN:0003-4975 [retrieved on Dec. 1, 2005]. cited by applicant

Dottori et al., "Echocardiographic imaging of the Alfieri type mitral valve repair," Ital. Heart J.,

2(4):319-320 (2001). cited by applicant

Downing et al., "Beating heart mitral valve surgery: Preliminary model and methodology," *Journal of Thoracic and Cardiovascular Surgery*, 123(6):1141-1146 (2002). cited by applicant

Extended European Search Report, dated Oct. 17, 2014, issued in European Patent Application No. 06751584.1. cited by applicant

Falk et al., "Computer-Enhanced Mitral Valve Surgery: Toward a Total Endoscopic Procedure," *Seminars in Thoracic and Cardiovascular Surgery*, 11(3):244-249 (1999). cited by applicant

Feldman et al, Randomized Comparison of Percutaneous Repair and Surgery for Mitral Regurgitation: 5-Year Results of Everest II. *J Am Coll Cardiol*. Dec. 29, 2015; 66(25):2844-2854.

cited by applicant

Fucci et al, Improved Results with Mitral Valve Repair Using New Surgical Techniques, *Eur. J. Cardiothorac. Surg.*, Nov. 1995, pp. 621-627, vol. 9. cited by applicant

Garcia-Rinaldi et al., "Left Ventricular Volume Reduction and Reconstruction in Ischemic Cardiomyopathy," *Journal of Cardiac Surgery*, 14:199-210 (1999). cited by applicant

Izzat et al., "Early Experience with Partial Left Ventriculectomy in the Asia-Pacific Region," *Annals of Thoracic Surgery*, 67:1703-1707 (1999). cited by applicant

Kallner et al., "Transaortic Approach for the Alfieri Stitch," *Ann Thorac Surg*, 71:378-380 (2001). cited by applicant

Kavarana et al., "Transaortic Repair of Mitral Regurgitation," *The Heart Surgery Forum*, #2000-2389, 3(1):24-28 (2000). cited by applicant

Khan et al., "Blade Atrial Septostomy; Experience with the First 50 Procedures", *Catheterization and Cardiovascular Diagnosis*, 23:257-262 (1991). cited by applicant

Kron et al., "Surgical Relocation of the Posterior Papillary Muscle in Chronic Ischemic Mitral Regurgitation," *Annals. Of Thoracic Surgery*, 74:600-601 (2002). cited by applicant

Kruger et al., "P73—Edge to Edge Technique in Complex Mitral Valve Repair," *Thorac Cardiovasc Surg.*, 48(Suppl. 1):106 (2000). cited by applicant

Maisano et al., "Valve Repair for Traumatic Tricuspid Regurgitation," *Eur. J. Cardio-Thorac Surg*, 10:867-873 (1996). cited by applicant

Maisano et al., "The Edge-to-edge Technique: A Simplified Method to Correct Mitral Insufficiency", *Eur. J. Cardiothorac. Surg.*, pp. 240-246, vol. 13 (Jan. 14, 1998). cited by applicant

Maisano et al., "The Double Orifice Repair for Barlow Disease: a Simple Solution for a Complex Repair," *Supplement I Circulation*, 100(18):1-94 (Nov. 1999). cited by applicant

Maisano et al., "The Hemodynamic Effects of Double-Orifice Valve Repair for Mitral Regurgitation: a 3D Computational Model," *European Journal of Cardio-thoracic Surgery*, 15:419-425 (1999). cited by applicant

Maisano et al., "The Double Orifice Technique as a Standardized Approach to Treat Mitral Regurgitation due to Severe Myxomatous Disease: Surgical Technique," *European Journal of Cardio-thoracic Surgery*, 17:201-205 (2000). cited by applicant

Maisano et al., "The Future of Transcatheter Mitral Valve Interventions: Competitive or Complementary Role of Repair vs. Replacement?", *Eur Heart J*.36(26):1651-1659 (Jul. 7, 2015). cited by applicant

Mantovani et al., "Edge-to-edge Repair of Congenital Familial Tricuspid Regurgitation: Case Report," *J. Heart Valve Dis.*, 9:641-643 (2000). cited by applicant

McCarthy et al, "Tricuspid Valve Repair With the Cosgrove-Edwards Annuloplasty System", *Ann. Thorac. Surg.*, 64:267-8 (Jan. 16, 1997). cited by applicant

McCarthy et al., "Partial Left Ventriculectomy and Mitral Valve Repair for End-Stage Congestive Heart Failure," *European Journal of Cardio-thoracic Surgery*, 13:337-343 (1998). cited by applicant

Moainie et al., "Correction of Traumatic Tricuspid Regurgitation Using the Double Orifice Technique," *Annals of Thoracic Surgery*, 73:963-965 (2002). cited by applicant

Morales et al., "Development of an Off Bypass Mitral Valve Repair," *The Heart Surgery Forum*

#1999-4693, 2(2):115-120 (1999). cited by applicant

Nakanishi et al., “Early Outcome with the Alfieri Mitral Valve Repair,” *J. Cardiol.*, 37: 263-266 (2001) [Abstract in English; Article in Japanese]. cited by applicant

Nielsen et al., “Edge-to-Edge Mitral Repair: Tension of the Approximating Suture and Leaflet Deformation During Acute Ischemic Mitral Regurgitation in the Ovine Heart,” *Circulation*, 104(Suppl. I):I-29-I-35 (2001). cited by applicant

Nishimura et al, 2014 AHA/ACC guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol.* Jun. 10, 2014; 63(22):2438-2488. cited by applicant

Osawa et al., “Partial Left Ventriculectomy in a 3-Year Old Boy with Dilated Cardiomyopathy,” *Japanese Journal of Thoracic and Cardiovascular Surg*, 48:590-593 (2000). cited by applicant

Park et al., *Clinical Use of Blade Atrial Septostomy*, *Circulation*, pp. 600-608, vol. 58, No. 4 (1978). cited by applicant

Patel et al., #57 Epicardial Atrial Defibrillation: Novel Treatment of Postoperative Atrial Fibrillation, 2003 STS Presentation, [Abstract Only]. cited by applicant

Privitera et al., “Alfieri Mitral Valve Repair: Clinical Outcome and Pathology,” *Circulation*, 106:e173-e174 (2002). cited by applicant

Redaelli et al., “A Computational Study of the Hemodynamics After ‘Edge-To-Edge’ Mitral Valve Repair,” *Journal of Biomechanical Engineering*, 123:565-570 (2001). cited by applicant

Reul et al., “Mitral Valve Reconstruction for Mitral Insufficiency,” *Progress in Cardiovascular Diseases*, XXXIX(6):567-599 (1997). cited by applicant

Rose et al., “Late MitraClip Failure: Removal Technique for Leaflet-Sparing Mitral Valve Repair”, *Journal of Cardiac Surgery*, (Jul 4, 2012), XP055047339, DOI: 10.1111/j. 1540-8191.2012.01483.x [retrieved on Nov. 11, 2012]. cited by applicant

Takizawa H et al: “Development of a microfine active bending catheter equipped with MIF tactile sensors”, *Micro Electro Mechanical Systems*, 1999. MEMS '99. Twelfth IEEE International Conference on Orlando, FL, USA Jan. 17-21, 1999, Piscataway, NJ, USA, IEEE, US, Jan. 17, 1999 (Jan. 17, 1999), pp. 412-417, XP010321677, ISBN: 978-0-7803-5194-3 figures 1-3. cited by applicant

Tamura et al., “Edge to Edge Repair for Mitral Regurgitation in a Patient with Chronic Hemodialysis: Report of A Case,” *Kyobu Geka. The Japanese Journal of Thoracic Surgery*, 54(9):788-790 (2001). cited by applicant

Timek et al., “Edge-to-edge mitral repair: gradients and three-dimensional annular dynamics in vivo during inotropic stimulation,” *Eur J. of Cardiothoracic Surg.*, 19:431-437 (2001). cited by applicant

Totaro, “Mitral valve repair for isolated prolapse of the anterior leaflet: an 11-year follow-up,” *European Journal of Cardio-thoracic Surgery*, 15:119-126 (1999). cited by applicant

Uchida et al., *Percutaneous Cardiomyotomy and Valvulotomy with Angioscopic Guidance*, *Am. Heart J.*, pp. 1221-1224, vol. 121 (Apr. 1991). cited by applicant

Umana et al., ‘Bow-Tie’ Mitral Valve Repair: An Adjuvant Technique for Ischemic Mitral Regurgitation, *Ann. Thorac. Surg.*, pp. 1640-1646, vol. 66 (May 12, 1998). cited by applicant

Votta et al., “3-D Computational Analysis of the Stress Distribution on the Leaflets after Edge-to-Edge Repair of Mitral Regurgitation,” *Journal of Heart Valve Disease*, 11:810-822 (2002). cited by applicant

U.S. Appl. No. 14/216,787, Apr. 8, 2016, Office Action. cited by applicant

U.S. Appl. No. 14/216,787, Nov. 7, 2016, Notice of Allowance. cited by applicant

U.S. Appl. No. 14/216,813, Mar. 9, 2017, Office Action. cited by applicant

U.S. Appl. No. 14/216,813, Dec. 15, 2017, Office Action. cited by applicant

U.S. Appl. No. 14/216,813, Apr. 6, 2018, Office Action. cited by applicant

U.S. Appl. No. 14/577,852, Oct. 20, 2016, Office Action. cited by applicant
U.S. Appl. No. 14/577,852, May 16, 2017, Office Action. cited by applicant
U.S. Appl. No. 14/577,852, Sep. 7, 2017, Office Action. cited by applicant
U.S. Appl. No. 14/577,852, Apr. 25, 2018, Notice of Allowance. cited by applicant
U.S. Appl. No. 15/423,060, Apr. 25, 2019, Office Action. cited by applicant
U.S. Appl. No. 15/423,060, Aug. 19, 2019, Office Action. cited by applicant
U.S. Appl. No. 15/423,060, Oct. 28, 2019, Office Action. cited by applicant
U.S. Appl. No. 15/423,060, Jan. 27, 2020, Notice of Allowance. cited by applicant
U.S. Appl. No. 15/642,245, Aug. 9, 2019, Office Action. cited by applicant
U.S. Appl. No. 15/642,245, Nov. 6, 2019, Notice of Allowance. cited by applicant
U.S. Appl. No. 15/642,245, Jan. 29, 2020, Notice of Allowance. cited by applicant
U.S. Appl. No. 15/642,245, Mar. 27, 2020, Notice of Allowance. cited by applicant
U.S. Appl. No. 15/724,545, Dec. 27, 2019, Office Action. cited by applicant
U.S. Appl. No. 15/724,545, May 1, 2020, Office Action. cited by applicant
Agricola et al., "Mitral Valve Reserve in Double Orifice Technique: an Exercise Echocardiographic Study," *Journal of Heart Valve Disease*, 11(5):637-643 (2002). cited by applicant
Alfieri, "The Edge-to-Edge Repair of the Mitral Valve," [Abstract] 6th Annual New Era Cardiac Care: Innovation & Technology, Heart Surgery Forum, (Jan. 2003) pp. 103. cited by applicant
Arisi et al., "Mitral Valve Repair with Alfieri Technique in Mitral Regurgitation of Diverse Etiology: Early Echocardiographic Results," *Circulation Supplement II*, 104(17):3240 (2001). cited by applicant
Bach et al, Early Improvement in Congestive Heart Failure After Correction of Secondary Mitral Regurgitation in End-stage Cardiomyopathy, *American Heart Journal*, Jun. 1995, pp. 1165-1170, vol. 129, No. 6. cited by applicant
Bach et al., Improvement Following Correction of Secondary Mitral Regurgitation in End-stage Cardiomyopathy With Mitral Annuloplasty, *Am. J. Cardiol.*, Oct. 15, 1996, pp. 966-969, vol. 78. cited by applicant
Bernal et al., "The Valve Racket: a new and different concept of atrioventricular valve repair," *Eur. J. Cardio-thoracic Surgery* 29:1026-1029 (2006). cited by applicant
Bhudia et al., "Edge-to-Edge (Alfieri) Mitral Repair: Results in Diverse Clinical Settings," *Ann Thorac Surg*, 77:1598-1606 (2004). cited by applicant
Bhudia, #58 Edge-to-edge mitral repair: a versatile mitral repair technique, 2003 STS Presentation, [Abstract Only], 2004. cited by applicant
Bolling et al., Surgery for Acquired Heart Disease: Early Outcome of Mitral Valve Reconstruction in Patients with End-stage Cardiomyopathy, *Journal of Thoracic and Cardiovascular Surgery*, Apr. 1995, pp. 676-683, vol. 109, No. 4. cited by applicant
Communication dated May 8, 2017, from the European Patent Office in counterpart European Application No. 04752714.8. cited by applicant
Dec et al, Idiopathic Dilated Cardiomyopathy, *The New England Journal of Medicine*, Dec. 8, 1994, pp. 1564-1575, vol. 331, No. 23. cited by applicant
Filsoufi et al., "Restoring Optimal Surface of Coaptation With a Mini Leaflet Prosthesis: A New Surgical Concept for the Correction of Mitral Valve Prolapse," *Intl. Soc. for Minimally Invasive Cardiothoracic Surgery* 1(4):186-87 (2006). cited by applicant
Frazier et al., #62 Early Clinical Experience with an Implantable, Intracardiac Circulatory Support Device: Operative Considerations and Physiologic Implications, 2003 STS Presentation, 1 page total. [Abstract Only]. cited by applicant
Fundaro et al., "Chordal Plication and Free Edge Remodeling for Mitral Anterior Leaflet Prolapse Repair: 8-Year Follow-up," *Annals of Thoracic Surgery*, 72:1515-1519 (2001). cited by applicant
Gateliene, "Early and postoperative results results of metal and tricuspid valve insufficiency surgical treatment using edge-to-edge central coaptation procedure," (Oct. 2002) 38 (Suppl 2):172

175. cited by applicant

Gatti et al., "The edge to edge technique as a trick to rescue an imperfect mitral valve repair," *Eur. J. Cardiothorac Surg*, 22:817-820 (2002). cited by applicant

Gundry, "Facile mitral valve repair utilizing leaflet edge approximation: midterm results of the Alfieri figure of eight repair," Presented at the Meeting of the Western Thoracic Surgical Association, (1999). cited by applicant

Gupta et al., #61 Influence of Older Donor Grafts on Heart Transplant Survival: Lack of Recipient Effects, 2003 STS Presentation, [Abstract Only]. cited by applicant

Ikeda et al., "Batista's Operation with Coronary Artery Bypass Grafting and Mitral Valve Plasty for Ischemic Dilated Cardiomyopathy," *The Japanese Journal of Thoracic and Cardiovascular Surgery*, 48:746-749 (2000). cited by applicant

International Search Report and Written Opinion dated Sep. 11, 2017 for International PCT Patent Application No. PCT/US2017/040977. cited by applicant

International Search Report and Written Opinion of PCT Application No. PCT/US2009/068023, dated Mar. 2, 2010, 10 pages total. cited by applicant

Kameda et al., Annuloplasty for Severe Mitral Regurgitation Due to Dilated Cardiomyopathy, *Ann. Thorac. Surg.*, 1996, pp. 1829-1832, vol. 61. cited by applicant

Kaza et al., "Ventricular Reconstruction Results in Improved Left Ventricular Function and Amelioration of Mitral Insufficiency," *Annals of Surgery*, 235(6):828-832 (2002). cited by applicant

Kherani et al., "The Edge-To-Edge Mitral Valve Repair: The Columbia Presbyterian Experience," *Ann. Thorac. Surg.*, 78:73-76 (2004). cited by applicant

Langer et al., "Posterier mitral leaflet extensions: An adjunctive repair option for ischemic mitral regurgitation?" *J Thorac Cardiovasc Surg*, 131:868-877 (2006). cited by applicant

Lorusso et al., "The double-orifice technique for mitral valve reconstruction: predictors of postoperative outcome," *Eur J. Cardiothorac Surg*, 20:583-589 (2001). cited by applicant

McCarthy et al., "Tricuspid Valve Repair With the Cosgrove-Edwards Annuloplasty System", *Ann. Throac Surg*. 64:267-8 (Jan. 16, 1997). cited by applicant

Noera et al., "Tricuspid Valve Incompetence Caused by Nonpenetrating Thoracic Trauma", *Annals of Thoracic Surgery*, 51:320-322 (1991). cited by applicant

Ricchi et al., Linear Segmental Annuloplasty for Mitral Valve Repair, *Ann. Thorac. Surg.*, Jan. 7, 1997, pp. 1805-1806, vol. 63. cited by applicant

Robicsek et al., #60 The Bicuspid Aortic Valve: How Does It Function? Why Does It Fail? 2003 STS Presentation, [Abstract Only]. cited by applicant

Supplemental European Search Report of EP Application No. 02746781, dated May 13, 2008, 3 pages total. cited by applicant

Supplementary European Search Report issued in European Application No. 05753261.6 dated Jun. 9, 2011, 3 pages total. cited by applicant

Tager et al, Long-Term Follow-Up of Rheumatic Patients Undergoing Left-Sided Valve Replacement With Tricuspid Annuloplasty—Validity of Preoperative Echocardiographic Criteria in the Decision to Perform Tricuspid Annuloplasty, *Am. J. Cardiol.*, Apr. 15, 1998, pp. 1013-1016, vol. 81. cited by applicant

Tibayan et al., #59 Annular Geometric Remodeling in Chronic Ischemic Mitral Regurgitation, 2003 STS Presentation, [Abstract Only]. cited by applicant

Timek, "Edge-to-Edge Mitral Valve Repair without Annuloplasty Ring in Acute Ischemic Mitral Regurgitation," [Abstract] *Clinical Science, Abstracts from Scientific Sessions*, 106(19):2281 (2002). cited by applicant

U.S. Provisional Application Filed on Jul. 6, 2016, by Khairkhahan., U.S. Appl. No. 62/359,121. cited by applicant

U.S. Provisional Application Filed on Nov. 7, 2016, by Khairkhahan., U.S. Appl. No. 62/418,571.

cited by applicant

U.S. Provisional Application Filed on Oct. 22, 2018, by Dale et al., U.S. Appl. No. 62/748,947.

cited by applicant

Umana et al., “‘Bow-tie’ Mitral Valve Repair Successfully Addresses Subvalvular Dysfunction in Ischemic Mitral Regurgitation,” Surgical Forum, XLVIII:279-280 (1997). cited by applicant

U.S. Appl. No. 14/216,813, filed Mar. 17, 2014, Hernandez. cited by applicant

Primary Examiner: Dang; Anh T

Attorney, Agent or Firm: Workman Nydegger

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION THE INVENTION (1) The present application is a divisional application of U.S. patent application Ser. No. 15/642,245, filed Jul. 5, 2017, now U.S. Pat. No. 10,736,632, which claims the benefit of provisional application No. 62/359,121, filed on Jul. 6, 2016, and of provisional application No. 62/418,571, filed on Nov. 7, 2016, the full disclosures of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. The Field of the Invention

(1) The present invention relates generally to medical devices and methods. More particularly, the present invention relates to devices, systems, kits, and methods for removing clips and other implanted prostheses from heart valves.

(2) Mitral regurgitation is a commonly encountered valvular disorder and prevalence increases with age. In this condition, blood regurgitates abnormally from the left ventricle into the left atrium during cardiac systole, and this condition can result in numerous adverse consequences such as heart failure due to left ventricular dysfunction, atrial fibrillation, pulmonary hypertension, and death. Published guidelines recommend surgical or transcatheter correction of mitral regurgitation to improve the clinical condition.

(3) Transcatheter correction of mitral regurgitation by implantation of a mitral valve clip (in particular the MitraClip® system from Abbott Vascular) has become a standard therapy for patients at high risk for open surgical corrective procedures. Such clip implantation procedures are performed through a guiding catheter that is inserted into the right femoral vein. One or more mitral valve clips can be delivered through the guiding catheter and implanted to re-approximate the anterior and posterior mitral leaflets (often referred to as an “edge-to-edge” repair). The MitraClip® mitral valve clip is metallic implant made of a cobalt chromium alloy and covered with a fabric mesh.

(4) To date, over 45,000 procedures using the MitraClip® valve clip have been performed globally and there are currently nearly 1000 procedures performed each month. The MitraClip® procedure has been found to be very safe procedure and provides a therapeutic option for patients at high surgical risk.

(5) There are, however, some significant limitations to conventional mitral valve clip technology. First, once a mitral valve clip is implanted, removal typically requires open surgical excision. Second, not all patients achieve satisfactory MR reduction at the time of the procedure due to technical challenges. Finally, up to 1 in 5 patients may have return of significant MR or have need for a repeat intervention within 6 months of the MitraClip® procedure.

(6) If patients have recurrent MR after a mitral valve clip procedure, current options for additional treatment are limited. One option would be to place another mitral valve clip but this is often not

possible due to concern for creating mitral stenosis (a mitral valve clip often narrows the valve so that the valve no longer opens properly).

(7) Other options include a variety of mitral valve repair and replacement technologies which are becoming available. Of great interest are transcatheter mitral valve replacement procedures where a bio-prosthetic mitral valve mounted on an expandable frame is deployed in a defective native mitral valve. Such transcatheter “replacement” valves offer complete elimination of mitral regurgitation and would mimic the surgical “gold standard” of mitral valve replacement in selected patients.

(8) At the present time, it can be difficult or impossible to implant a transcatheter mitral valve when a prior mitral valve clip procedure has been performed, since the mitral valve clip(s) interfere and will not allow complete expansion of the new valve. It would therefore be desirable to provide devices, systems, kits, and methods for the transcatheter removal of clips and other implanted prostheses from heart valves. It would be particularly desirable that such devices, systems, kits, and methods for the transcatheter removal of clips and other implanted prostheses from heart valves would leave the heart valve in condition to receive a subsequent transcatheter procedure, such as prosthetic valve implantation, to treat the valve pathology. At least some of these objectives will be met by the inventions described herein.

2. Description of the Background Art

(9) See US Patent Publs. 2014/0228871; US2015257883; and US2014135799; U.S. Pat. Nos. 8,500,768; 7,955,340; and 5,895,404; as well as the following guidelines and publications: Nishimura R A, Otto C M, Bonow R O et al. 2014 AHA/ACC guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol 2014; 63:2438-88; Feldman T, Kar S, Elmariah S et al. Randomized Comparison of Percutaneous Repair and Surgery for Mitral Regurgitation: 5-Year Results of EVEREST II. J Am Coll Cardiol 2015; 66:2844-54; and Maisano F, Alfieri O, Banai S et al. The future of transcatheter mitral valve interventions: Competitive or complementary role of repair vs. replacement?, Eur Heart J 2015; 36:1651-9.

SUMMARY OF THE INVENTION

(10) The present invention provides transcatheter tools and procedures that can be used to separate and optionally remove one or more mitral valve clips from native mitral leaflet tissue. In such procedures, the physician can insert specifically designed catheters either from the left atrium or left ventricle. A guidewire may be placed around the leaflet and valvular tissue adjacent to the previously implanted mitral valve clip. Utilizing mechanical cutting or abrasion, or radiofrequency energy, the mitral valve clips may be excised from either the anterior or posterior leaflets, or both, restoring the native valve orifice and allowing future repair or replacement therapy. After separation of the mitral valve clips from native leaflet tissue, the degree of mitral regurgitation would be expected to worsen acutely. The separation procedure of the present invention in contrast can be performed immediately prior to transcatheter valve implantation or another transcatheter corrective procedure. For instance, if transapical access were used for deployment of a new transcatheter mitral valve, then transseptal access could be used to release the mitral valve clips, or vice versa.

(11) In a first aspect of the present invention, a method for excising a clip approximating an opposed pair of valve leaflets in a heart valve comprises introducing a capture catheter into a heart chamber adjacent to the valve leaflets on one side of the clip. A transfer catheter is also introduced into the heart chamber adjacent to the valve leaflets on another side of the clip, and a cutting member is deployed between the transfer catheter and the capture catheter to place or engage the cutting member against tissue of at least one of the valve leaflets. The cutting member is then used to excise the clip from at least one of the valve leaflets to release fixation and allow separation of the valve leaflets.

(12) The excision methods of the present invention are useful on any heart valve where a clip may have been placed to enhance coaptation or for other reasons. Most typically, the heart valve is a mitral valve, and the catheters may be introduced transeptally into the left atrium above the valve or may be introduced transapically into the left ventricle beneath the mitral valve. In other instances, the target heart valve can be a tricuspid valve, and the catheters may be introduced into the right atrium above the tricuspid valve or into the right ventricle below the tricuspid valve.

(13) In exemplary embodiments, deploying the cutting member comprises steering a tip of at least one and usually both of the capture catheter and the transfer catheter to bring the tips of both catheters into proximity with the valve clip. Typically, at least one of the capture catheter and transfer catheter may be advanced through an opening between the valve leaflets adjacent to the fixed valve clip, where the advanced catheter may then be steered around the opposite side of the valve clip and brought up through the space between the valve leaflets on an opposite side of the valve clip. The cutting member may then be passed between the transfer catheter and the capture catheter to span a target region of the valve leaflet to be excised. The catheters may then be manipulated to extend the cutting member past a side of the valve clip so that the cutting member is positioned to excise a portion of the valve leaflet on one side of the valve. As will be described in more detail below, the valve clip may be left implanted in the opposed valve leaflet, and a subsequent valve intervention procedure performed with the clip left attached to the opposed valve leaflet. Alternatively, the capture and transfer catheters may be used to reposition the cutting member on the other side of the valve clip so that the opposed leaflet may be excised from the clip to completely release the clip and allow the clip to be removed from the heart prior to performing a subsequent intervention on the heart valve. When releasing and removing the valve clip, it will of course be necessary that the valve clip be constrained, typically by a removal catheter as described herein below.

(14) In still further specific examples, the cutting member may be deployed by engaging a magnetic element on the capture catheter against a magnetic element on the cutting member. The magnetic elements are typically at the tip of the capture catheter and an end of the cutting member, and the catheter(s) may be manipulated to draw the cutting member past either or both sides of the valve clip in order to excise of the valve leaflet tissue to release the clip.

(15) In an alternative embodiment, the cutting element may comprise a loop which is used to capture a free end of the cutting member, and the capture catheter may then be manipulated to advance the loop over a free end of the cutting member and draw the cutting member past the clip through the valve tissue.

(16) In still further embodiments, deploying the cutting member may comprise drawing the tips of the capture catheter and the transfer catheter together to form a path past the valve clip and advancing the element over the path. After such advancement and positioning of the cutting member, the capture catheter and/or the transfer catheter may be manipulated to manipulate the cutting member to excise the valve tissue to release the clip.

(17) As discussed above, in some embodiments of the methods of the present invention, the clip(s) will be excised from one valve leaflet only, leaving the clip(s) implanted in an opposed valve leaflet. Freeing of the clip(s) from at least one valve leaflet will in at least some cases be sufficient to release the opposed valve leaflets and enable performance of the subsequent valve replacement or other interventional procedure. In other instances, however, it will be preferable to completely remove the valve clip(s) from the heart valve prior to performing a subsequent valve implantation or other intervention. In such cases, a removal catheter or other device will typically be used to constrain or capture the clip(s) during the excision and withdraw the excised clip(s) from the heart chamber prior to any further intervention.

(18) In still other specific embodiments of the methods of the present invention, the clip(s) may be excised from the valve leaflet tissue in a variety of ways. For example, the cutting member may comprise an electrode segment or other electrically conductive region capable of delivering a

radiofrequency (RF) cutting current to the tissue. In particular, the RF current would be delivered in a cutting mode so that the cutting member could excise the valve tissue adjacent to the valve clip. In other instances, the cutting member could comprise a sharpened or abrasive region which could be used to mechanically cut through the valve tissue. For example, a sharpened or abrasive cutting region could be reciprocated so that it would “saw” through the tissue in order to excise the valve. Other known tissue cutting modalities could also be employed.

(19) As described thus far, the capture catheter, transfer catheter, and cutting member are used to excise the tissue from a single valve leaflet, leaving the valve clip attached to the opposed valve leaflet. While removing the valve from a single leaflet will allow sufficient opening of the leaflets for subsequent prosthetic valve implantation or other corrective procedures, it will sometimes be desirable to remove the valve clip from both opposed valve leaflets. In such instances, it will be preferred to introduce a clip removal catheter into the heart chamber in order to stabilize the eventually remove the valve clip. A distal end of the clip removal catheter can be engaged against the clip and will usually be attached to the clip while the clip is excised from one or both valve leaflets using the cutting member.

(20) In still further exemplary embodiments, the capture catheter, the transfer catheter, and optionally the clip removal catheter can be introduced transseptally, usually simultaneously through a transseptal catheter or sheath. In still other specific embodiments, the capture catheter, the transfer catheter, and optionally the clip removal catheter can be introduced transapically, typically simultaneously through a transapical sheath or catheter.

(21) In a second aspect of the present invention, a system for excising an implanted clip approximating opposed valve leaflets in the heart valve comprises a capture catheter, a transfer catheter, and a cutting member. The capture catheter is configured to be introduced into a heart chamber adjacent to the valve leaflets on one side of the clip. The transfer catheter is configured to be introduced into the heart chamber adjacent to the valve leaflets on another side of the clip, and the cutting member is configured to be deployed from the transfer catheter to the capture catheter to place the cutting element against tissue of at least one of the valve leaflets to excise the clip.

(22) In specific aspects of the system, at least one of the capture catheter and the transfer catheter will have a steerable tip, usually both having a steerable tip. The capture catheter will often have a magnetic distal tip configured to engage and capture a magnetic element disposed at the distal end of the cutting member. In this way, the magnetic tip on the capture catheter can be used to attract and engage the magnetic end of the cutting member so that the cutting member can be deployed between the capture catheter and the transfer catheter. Thus, a cutting region of the cutting member can be positioned to engage the valve leaflet tissue adjacent to the valve clip so that the cutting region can be used to cut the tissue and excise the clip. For example, the cutting region could comprise a sharpened or abrasive region which can be used to mechanically cut the valve tissue. Alternatively, the cutting region could comprise an RF electrode which can be powered with a cutting current to excise the valve leaflet tissue adjacent to the clip.

(23) The systems of the present invention may further comprise an introductory sheath, typically a transseptal sheath or a transapical sheath, for delivering the catheters of the present invention to a heart chamber. Usually, the introductory sheath will be sufficiently large to accommodate at least the capture catheter and the transfer catheter simultaneously. Often times, the system will further comprise a removal catheter, and the introductory sheath will be sized to be sufficiently large to simultaneously accommodate each of the capture catheter, transfer catheter, and removal catheter.

(24) The clip removal catheter of the present invention will typically have a distal end configured to engage and capture the clip. Certain valve clips, such as the MitraClip® valve clip, have a unique shape that provides a “docking” feature which can be used by the clip removal and optionally other catheters of the present invention to “dock” with either the atrial or ventricular aspect of the valve clip. The shape of the valve clip(s) is typically easy to discern on fluoroscopy which facilitates targeting and optionally docking a removal or other catheter with the clip during a removal

procedure. Such docking has several advantages. First, docking can stabilize the valve clip relative to the catheter system which is advantageous as the valves are moving during the beating heart procedure. Second, docking facilitates orienting the catheter(s) to improve position of the cutting member, e.g. looping a cutting wire so that it encircles the clip using either the magnetic or snare techniques shown above. Third, by docking a removal or other catheter with the valve clip, the catheter and cutting member can be stably positioned in close proximity to the valve clip, thus minimizing the risk of entangling or snaring the chordal apparatus with the loop wire or other cutting member. Fourth, docking can be performed either from an atrial or a ventricular aspect. For example, atrial docking can be achieved by docking into a groove on the top of a “Y” portion of the MitraClip® clip. Ventricular docking may be achieved by docking with the bottom of the “Y” portion of the MitraClip® clip. The docking can be passive, e.g. via an interference fit, or can be active where the removal or other catheter has a “grasping” feature to securely attach the MitraClip® clip. Docking can be performed on each valve clip individually or on several at the same time with modification in the docking feature.

(25) The removal catheters of the present invention may be designed to dock with the MitraClip® or other valve clips to place a snare around the clip. After the clip is freed from both the anterior and posterior leaflets (or septal leaflet in case of a tricuspid valve) using the previously described catheters, the clip can be removed from the body through the access sheath by the removal catheter. The removal catheter may have any one of a variety of clip-grasping features, such as a snare, biopsy-type forceps, or jaws that secure the valve clip to hold it after it has been freed from both leaflets so it can be removed from the body. The capture and/or transfer catheter may also be configured to dock with the valve clip, e.g. have a distal tip with a shape complementary to the shape of the valve clip. The catheter that removes the clip can be placed via a transvenous transseptal approach or left ventricular transapical approach. In some instances, a removal catheter can be designed with a funnel shaped tip to allow the valve clip to be drawn into the catheter with proper orientation.

(26) The catheter systems of the present invention can also be used to encircle and remove pacemaker or defibrillator leads. For example, by sliding a loop encircling the lead along a length of the lead, the lead can be excised from the heart or vascular tissue in which it has been implanted. In particular, the catheters of the present invention can be used to free leads from vascular venous tissue such as the axillary vein, brachiocephalic vein, and/or superior vena cava. The catheters of the present invention can also be used to free leads from a tricuspid valve or the right ventricular myocardium. In particular, a lead may be freed from the tricuspid valve in situations where the lead may be causing tricuspid regurgitation (TR). The lead could optionally be repositioned to reduce TR. The catheters can also be used to free a lead to allow the lead to be repositioned in the tricuspid valve orifice as desired to facilitate placement of a transcatheter valve in the tricuspid position.

(27) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying drawings, wherein:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1 and 1A illustrate the anatomy of the mitral valve after implantation of a mitral valve clip.

(2) FIGS. 2A and 2B illustrate a capture catheter constructed in accordance with the principles of the present invention with a steerable distal region shown in its straightened configuration (FIG. 2A) and in deflected configurations (FIG. 2B).

- (3) FIG. 3 illustrates a transfer catheter constructed in accordance with the principles of the present invention.
- (4) FIG. 4 illustrates a cutting member used with the capture and transfer catheters in the methods of the present invention.
- (5) FIGS. 5A through 5E illustrate different embodiments of magnetic tips on the capture catheters and magnetic distal elements on the cutting members.
- (6) FIG. 6 illustrates an alternative system according to the present invention including a capture catheter having a capture loop at its distal end.
- (7) FIGS. 7A through 7F illustrate use of a capture catheter and a transfer catheter in a method for removing a pair of clips implanted in a mitral valve in accordance with the principles of the present invention.
- (8) FIG. 8 illustrates transapical introduction of a capture catheter and a transfer catheter into a left atrium for excising a valve clip in a mitral valve in accordance with the principles of the present invention.
- (9) FIGS. 9A through 9C illustrate use of a capture catheter and a transfer catheter in a method for removing a valve clip from a tricuspid valve in accordance with the principles of the present invention.
- (10) FIGS. 10A through 10E illustrate use of a capture catheter and a transfer catheter in a method for removing a single valve clip from a mitral valve in accordance with the principles of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(11) The present invention provides apparatus and methods for the separation of one or more valve clips from mitral and other heart valve leaflets. The valve clips can be located anywhere along the valve coaptation plane (central, medial, lateral, or commissural), and in some embodiments the valve clips are removed from one of the coapting valve leaflets while being left in place in the other of the coapting valve leaflets. In other embodiments, the valve clips will be removed from both valve leaflets and be extracted entirely from the heart.

(12) The technology of the present invention can be deployed from a left atrial aspect (transseptal or direct atrial access), or from the left ventricular aspect (transapical, direct ventricular puncture, or retrograde aortic access). The cutting action can be provided by a “noose,” by a scissor-like device, by a wire-based cutter similar to a “cheese cutter,” by a radiofrequency electrode cutting element, or the like. In some instances, the mitral valve clips may be removed from both anterior and posterior leaflets, and a snare, biopptome-type device, or other removal catheter may be used to completely remove the mitral valve clip from the body. The present invention may be used to remove valve clips from tricuspid valves as well as mitral valves, and the catheters be placed through transseptal and transapical guiding catheter which are typically large enough to accommodate all catheters being used in the procedure simultaneously.

(13) Referring now to FIGS. 1 and 1A, valve clips C may be implanted in the leaflets of a mitral valve MV as well as a tricuspid valve TV in a heart H. The mitral valve MV separates the left atrium LA from the left ventricle LV, and the tricuspid valve TV separates the right atrium RA from the right ventricle RV. For completeness, it is noted that blood flows from the left ventricle LV through the aortic valve AV into the aorta A. As shown in more detail in FIG. 1A, the clip C in the mitral valve MV may be located near a middle of the valve opening between the anterior leaflet AL and the posterior leaflet PL. The clip C, however, could also be implanted at other locations between the leaflets away from the middle of the valve opening.

(14) Referring now to FIGS. 2A and 2B, a capture catheter 10 comprises a shaft 12 having a proximal end 14 and a distal end 16. A control handle 18 is attached to the proximal end of the shaft, and a deflection knob 20 is located at the distal of the handle. The capture catheter 10 will typically also include a flush port 22 and will have a central lumen adapted to receive a guidewire 24 having a shapeable tip 26 at its distal end. As shown in particular FIG. 2B, a distal region 28 of

the catheter shaft **12** will be actively deflectable or “steerable” so that the region can be deflected by up to 180°, as shown. Such deflection will be controlled by the deflection knob **20**, and suitable deflection mechanisms may include pull wires, slotted regions at the distal end of the shaft, or any other conventional catheter deflection technology.

(15) Referring now to FIG. 3, a transfer catheter **34** includes a shaft **36** having a proximal end **38** and a distal end **40**. A control handle **42** is attached to the proximal end of the shaft, and a deflection knob **44** is disposed at the distal end of the control handle **42**. The transfer catheter **34** may also include a flush port **46** and will typically have a steerable distal tip **48** similar to that described above for the capture catheter **10**.

(16) Referring now to FIG. 4, a cutting member **50** comprises an elongate body **52** which typically has a wire-like configuration (typically being electrically conductive in the case of electrosurgical cutting members as described below) so that it may be fed through an advancement lumen of the transfer catheter **34**. The cutting member **50** further includes a cutting region **54** which typically spans a region near a middle of the length of the elongate body, typically having a length from 0.5 cm to several centimeters, and which typically is bounded on each side by a radiopaque marker **56**. In the case of electrosurgical cutting members, the member may comprise an electrically conductive wire where the cutting region is uninsulated and the regions on either side of the cutting region are electrically insulated. A magnetic distal element **58** will typically be located at one end of the elongate body **52** and, as shown in FIG. 3, when the cutting member **50** is loaded onto the transfer catheter **34**, the magnetic distal element **58** will extend outwardly from the distal tip **40** of the catheter so that it is exposed and available to magnetically attract the magnetic tip **30** of the capture catheter **10** as described below.

(17) As shown in FIGS. 5A through 5E, the magnetic tip **30** of the capture catheter **10** may take a variety of forms. Magnetic tip **30a** (FIG. 5A) has a bullet shape with a through lumen to accommodate the guidewire of the capture catheter. Magnetic tip **30b** has a disc-like shape and also has a through lumen to accommodate the guidewire in the capture catheter.

(18) The magnetic distal element **58** of the cutting member **50** may also have a variety of configurations. Most simply, the magnetic distal element **58** will have a sphere shape as shown in FIG. 5E. Alternatively, the magnetic distal element may have a bullet shape **58a** as shown in FIG. 5B or a disc-like shape **58b** as shown in FIG. 5D.

(19) As described thus far, the capture catheter **10** and the cutting member **50** will each have magnetic elements to allow capture of the cutting member by the capture catheter **10**. While this is a preferred design, a variety of other capture mechanisms could also be employed. For example, as shown in FIG. 6, a capture catheter **62** may carry a capture loop **66** at its distal end. The capture loop could be used then to capture a free end **70** of a cutting member **72** carried by transfer catheter **64**. The use of hooks, barbs, coil targets, and the like, would also be possible to allow for capture of the cutting member by a capture catheter.

(20) Referring now to FIGS. 7A through 7F, the use of capture catheter **10** and transfer catheter **34** for removing a pair of clips C1 and C2 from a mitral valve MV via a transseptal approach will be described. A steerable distal tip **28** of guidewire **24** is first introduced through a transseptal sheath **74** into a left atrium above the mitral valve. The tip **28** can be advanced between the clips C1 and C2 under fluoroscopic guidance with the heart beating. Optionally, the capture catheter **10** could be used to assist in placing the guide wire **24** between the clips before the catheter **10** is advanced over the guide wire. The guide wire can be passive or active.

(21) After the guidewire is placed, as shown in FIG. 7A, the capture catheter **10** can be advanced over the guidewire **24** so that it is positioned above the clips C1 and C2, as shown in FIG. 7B. The distal end of the capture catheter **10** is then advanced between the clips C1 and C2 so that the magnetic tip enters the left ventricle, as shown in FIG. 7C. The transfer catheter **34** may then be introduced through the transseptal sheath **74** in parallel to the capture catheter **10** and steered to advance the magnetic distal element **58b** on the cutting member **50** so that it engages the magnetic

tip **30** on the capture catheter **10**. For example, the steerable distal region **48** of the transfer catheter **34** can be advanced downwardly between the anterior leaflet AL and posterior leaflet PL and manipulated to engage the magnetic distal element **58b** against the magnetic tip **30**.

(22) After the cutting member **50** and capture catheter **10** are coupled together, the capture catheter **10** may be pulled back to expose the cutting region **54**, and the catheters manipulated to engage the cutting region **54** against the second valve clip C2, as shown in FIG. 7E. The cutting region **54** can then be manipulated to saw through or electrode surgically cut the tissue of the posterior leaflet PL surrounding the clip C2. The procedure may then be repeated to excise the first clip C1 from the posterior leaflet PL, as shown in FIG. 7F, resulting in opening of the mitral valve to allow a prosthetic valve implantation or other intervention to take place. While excision of the posterior leaflet is illustrated in FIGS. 7A-7F, in some instances it would be preferable to excise the anterior leaflet tissue leaving the clip(s) attached to the posterior leaflet.

(23) Referring now to FIG. 8, excision of valve clip C from the mitral valve MV can also be performed using capture catheter **10** and transfer catheter **34** through a transapical sheath **78**. Using the transapical approach, the distal tips of the catheters **10** and **34** will be advanced upwardly from the left ventricle LV to the left atrium LA, and the tissue can then be excised using the catheters as generally as described previously.

(24) Referring now to FIGS. 9A through 9C, excision of a valve clip C implanted between an anterior leaflet AL and a septal leaflet SL in a tricuspid valve TV will be described. Capture catheter **10** and transfer catheter **34** can be introduced through an access sheath **80** into the right atrium above the tricuspid valve, as shown in FIG. 9B. The catheters **10** and **34** can be used to position the cutting region **54** of the cutting member **50** on a side of the clip C in the septal leaflet SL as shown in FIG. 9C. The cutting member **50** can then be manipulated in any of the ways described previously to excise the tissue of the septal leaflet SL to release the clip C and open the tricuspid valve, as shown in FIG. 9C. The portion of the clip C implanted in the anterior leaflet can also be excised, or the clip can be left in place, prior to performing a subsequent intervention on the valve.

(25) Referring now to FIGS. 10A through 10E, a single valve clip C can be removed from a mitral valve MV by introducing the capture catheter **10** and the transfer catheter **34** into the left atrium using a transseptal sheath **74**, as generally described previously. In addition, a clip removal catheter **90** can be introduced through the transseptal sheath **74** simultaneously with the capture catheter **10** and the transfer catheter **34** in order to engage and stabilize the clip C. As shown in FIG. 10A, a clip capture element **94** at the distal end of a manipulation wire **92** can be delivered through a lumen of the clip removal catheter **90** to capture an upper surface of the clip C. The cutting region **54** of the cutting member **50** is then used to excise the tissue in the anterior leaflet AL, while the clip remains held and stabilized by the clip removal catheter **90**, thus freeing the clip from the anterior leaflet, as shown in FIG. 10B (where the capture catheter **10** and transfer catheter **34** are removed for ease of illustration). The capture catheter **34** and transfer catheter **10** may then be repositioned, as shown in FIG. 10C, so that the cutting region **54** of the cutting member **50** is positioned against the posterior leaflet PL, and the cutting region can then be manipulated to excise tissue in the posterior leaflet to release the clip, as shown in FIG. 10D. The clip removal catheter **90**, which continues to hold the clip C, can be then be used to withdraw the clip C from the mitral valve, as shown in FIG. 10E. The mitral valve MV is then completely free from implanted clips to receive a mitral valve prosthesis or to have another corrective intervention performed.

(26) Although certain embodiments of the disclosure have been described in detail, certain variations and modifications will be apparent to those skilled in the art, including embodiments that do not provide all the features and benefits described herein. It will be understood by those skilled in the art that the present disclosure extends beyond the specifically disclosed embodiments to other alternative or additional embodiments and/or uses and obvious modifications and equivalents thereof. In addition, while a number of variations have been shown and described in varying detail,

other modifications, which are within the scope of the present disclosure, will be readily apparent to those of skill in the art based upon this disclosure. It is also contemplated that various combinations or sub-combinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the present disclosure. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the present disclosure. Thus, it is intended that the scope of the present disclosure herein disclosed should not be limited by the particular disclosed embodiments described above. For all of the embodiments described above, the steps of any methods need not be performed sequentially.

Claims

1. A system for excising an implanted clip approximating opposed valve leaflets in a heart valve, said system comprising: a capture catheter configured to be introduced into a heart chamber adjacent to the valve leaflets on one side of the clip; a transfer catheter configured to be introduced into the heart chamber adjacent to the valve leaflets on another side of the clip; and a cutting member configured to be deployed from the transfer catheter to the capture catheter to place the cutting member against tissue of at least one of the valve leaflets to excise the clip, wherein the cutting member comprises a cutting zone formed of an uninsulated conductive member bounded by a first radiopaque marker and a second radiopaque marker.
2. A system as in claim 1, wherein at least one of the capture catheter and the transfer catheter has a steerable distal tip.
3. A system as in claim 1, wherein each of the capture catheter and the transfer catheter has a steerable distal tip.
4. A system as in claim 1, wherein the capture catheter has a magnetic distal tip.
5. A system as in claim 4, wherein the cutting member is carried by the transfer catheter and has a magnetic element exposed at a distal end of the transfer catheter so that manipulation of the transfer catheter positions within the heart chamber allows the capture catheter to capture the cutting member.
6. A system as in claim 4, wherein the magnetic distal tip has one of a sphere shape, a bullet shape, or a disc-like shape.
7. A system as in claim 1, further comprising a clip removal catheter having a distal end configured to engage and capture the clip.
8. A system as in claim 1, wherein the cutting member comprises a radiofrequency electrode for energized cutting of valve tissue.
9. A system as in claim 1, wherein the cutting member comprises a mechanically abrasive or sharpened edge for mechanically cutting valve tissue.
10. A system as in claim 1, further comprising a delivery sheath configured to introduce the capture catheter and the transfer catheter to the heart chamber.
11. A system as in claim 1, wherein a delivery sheath has a lumen sized to simultaneously accommodate both the capture catheter and the transfer catheter.
12. A system as in claim 1, wherein a delivery sheath comprises a transseptal sheath.
13. A system as in claim 1, wherein a delivery sheath comprises a transapical sheath.
14. A system for excising an implanted clip approximating opposed valve leaflets in a heart valve, said system comprising: a capture catheter configured to be introduced into a heart chamber adjacent to the valve leaflets on one side of the clip, the capture catheter comprising a magnetic tip; a transfer catheter configured to be introduced into the heart chamber adjacent to the valve leaflets on another side of the clip; and a cutting member configured to be deployed from the transfer catheter to the capture catheter to place the cutting member against tissue of at least one of the valve leaflets to excise the clip, the cutting member comprising a magnetic element exposed at a

distal end of the transfer catheter during delivery so that manipulation of the transfer catheter positions the magnetic element within the heart chamber and allows magnetic attraction of the magnetic tip and the magnetic element so the capture catheter contacts a portion of the cutting member, wherein the cutting member comprises a cutting zone formed of uninsulated conductive member and bounded by a first radiopaque marker and a second radiopaque marker.

15. A system as in claim 14, further comprising a transseptal sheath configured to receive the capture catheter and the transfer catheter.

16. A system as in claim 15, wherein the cutting zone includes a radiofrequency electrode, an abrasive region, or a sharpened edge for cutting valve tissue.

17. A system as in claim 16, further comprising a guidewire with a shapeable tip, the capture catheter being configured to be advanced along the guidewire to the heart chamber.

18. A system as in claim 17, further comprising a clip removal catheter configured to be introduced into a heart chamber adjacent to the valve leaflets.

19. A system as in claim 18, further comprising a clip capture element configured to be selectively delivered through a lumen of the clip removal catheter to capture the clip.

20. A system for excising an implanted clip approximating opposed valve leaflets in a heart valve, said system comprising: a capture catheter configured to be introduced into a heart chamber adjacent to the valve leaflets on one side of the clip; a transfer catheter configured to be introduced into the heart chamber adjacent to the valve leaflets on another side of the clip; and a cutting member configured to be deployed from the transfer catheter to the capture catheter to place the cutting member against tissue of at least one of the valve leaflets to excise the clip, wherein the cutting member comprising at least one selected from a group of a sharpened member, an abrasive member, and an RF electrode member, wherein the cutting member comprises an elongate cutting zone formed of an uninsulated conductive member proximate a distal end of the cutting member, the cutting zone bounded by a first radiopaque marker and a second radiopaque marker.

21. A system as in claim 20, wherein the cutting zone is located near a middle portion of the cutting member.
